



Variable impedance models including fuzzy fractional order for control of human–robot interaction: a systematic review

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Abstract

In Industry 4.0 and 5.0, human–robot interaction (HRI) is critical and common due to safety, reliability, high performance, and productivity speed up process while maintaining high productivity. The development in HRI systems led to further classifications such as human–robot collaboration (HRC) that is a fundamental principle in production environment. However, the implementations of HRC required critical considerations such as understanding the main aim of the collaborations, the movability, and dynamics in the working environment that specific types of controller model can be used and avoiding undesirable collisions with humans. Since selecting the accurate type of controller is essential for accomplishing the required accuracy with optimum performance and minimum error with handling complexity especially in delicate systems, this systematic review investigates and permits an intensive insight into impedance controller in HRI/HRC systems, with special focus on progress of variable impedance model, mainly by using fractional-order and fuzzy (*I* and *II*) methods. Besides, hybrid approaches in which learning-based methods are implemented are also highlighted. The differences between linear and nonlinear models are also pointed. Though, various industrial applications were discussed to emphasize the importance of cobots in innovation industries. It is observed that in dominant, the interaction force relying on impedance model was considered a linear structure with unknown coefficient whether the system is linear or nonlinear, and the adopted learning methods were only responsible to define the optimal (or near optimal) values of the linear components adopted in those models.

Keywords Human–robot interaction · Impedance control · Variable impedance · Fractional order interaction model · Fuzzy control

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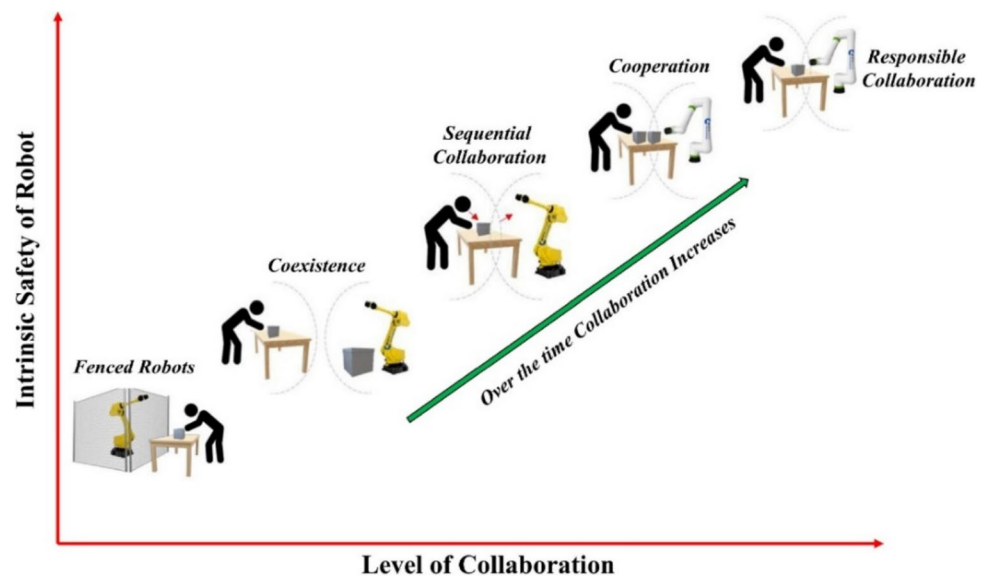
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1 Introduction

In the recent decade, significant interests and efforts have shown into developing human–robot interaction (HRI) [1]. Besides, human–robot collaboration (HRC) in working environments has become a key component in the development programs of large industries and has been realized in many well-known industrial factories worldwide; see Fig. 1 for various types of sharing space in HRC system [2, 3]. Although, the implementation of robots especially robotic arms inside several factories were common for a few decades, typically the operated in preserved and isolated environments separated from humans [4]. However, what has emerged today is direct physical or intellectual collaboration and interaction of humans and robots in common working environments to maximize productivity [5]. Besides, the presence of robots in common working environments to cooperate with humans (cobots) provides new

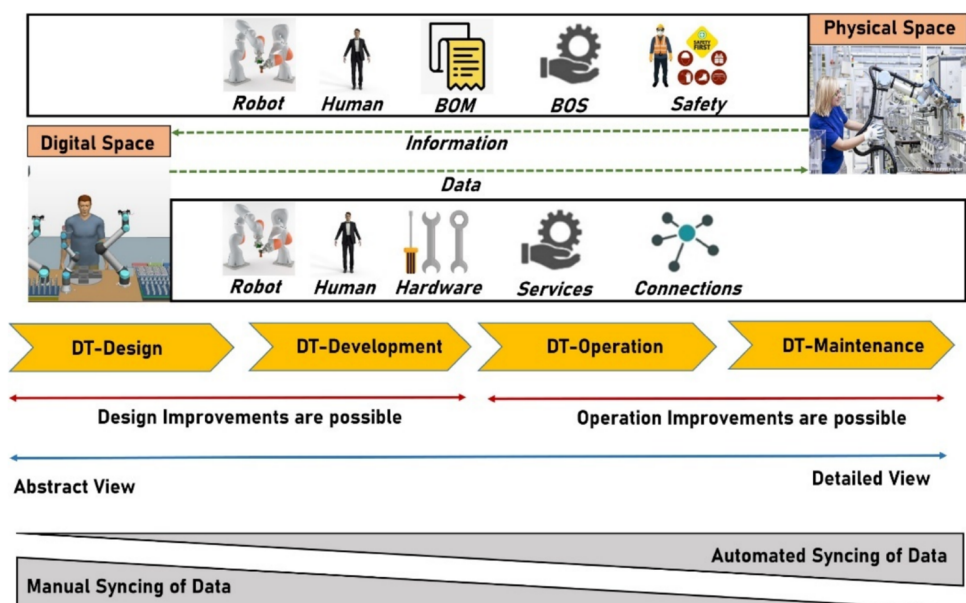
Fig. 1 Various types of shared work space as cobots in HRC system [2]



opportunities and prevents various challenges [6]. Also, it exploits developing control methodologies, improving significant flexibility and productivity, while avoiding worker fatigue and injuries [7]. In relevant literature, the issue of safety in common robot-human working environments has attracted great attention and HRI levels have been defined accordingly [8]. Furthermore, the safety along this interaction is essential and highlighted in literature review [9, 10], where these cobots cause to boost the quality in manufacturing sector by 15% and minimize the assembly time by 30% [11]. However, over the significant desirable advantages of HRC system, it might still face some challenges, such as safety and work-force concerns, cost, skills gaps, complexity, etc. This challenges can be overcome by implementing

the digital twins (DTs) in HRC systems [12]. The DTs are defined as virtual duplication of a physical system or an object [13]. Figure 2 illustrates the implementations of DTs in HRC system [2]. The industrial ecosystems can be redefined by incorporation of cobots, DTs, collaborative intelligence (CI) encourage resilient, personalized, and smart manufacturing, where such developed techniques improve human capabilities instead of changing them, confirming an efficient, sustainable, and adaptive production model [7]. As an example of implementing DTs with industrial robots, Liu et al. [14] developed a DT-assist optimization framework for fictional-order PID (FOPID) controller. Meanwhile, in parallel with developed technologies in HRI and various control approaches, the operational reliability of robotic systems in

Fig. 2 Implementation of DTs in HRC system [2]



industries is considered the main critical concern. Method such as fault detection and diagnosis (FDD) help the DTs and artificial intelligence (AI) to enhance safety and reliability by reducing downtime in HRI/HRC environment. Malik and Brem [15] developed a DT for HRI tasks for the purpose of allowing predictive control and parameter tuning without implementing direct learning methodologies. The DT supports risk estimation, advanced monitoring, and adaptive scheduling. However, since fifth industrial revolution is actively integrating human–machine collaboration in various manufacturing sectors, it causes the cobots systems to be further empowered in Industry 5.0 rather than Industry 4.0 [16]. Besides, this collaboration is considered the heart of Industry 5.0.

However, various methods have been developed for the interaction of a robotic manipulator with the surrounding including human, and the majority of these methods were found to rely on force control [17]. Furthermore, the impedance control is a unique control method to obtain the desired dynamic balance between robotic manipulator and environment [18, 19]. Especially, in HRI, the environment of robot is human and the force exerted from human to robot wrist end effector (EE) is measured by torque/force sensor. Nonetheless, in impedance control, the aim is to control a dynamic relationship to determine the requirements of a task which involves physical interaction with human. Another challenge here is “desired interaction force” which is diverse since the accomplished tasks vary and finding the desired dynamic for this interaction force is an additional problem. Therefore, in relevant literature, variable impedance control (VIC) has been introduced and developed.

For centuries, humankind has aspired for creating and developing a machine which can act and perform as human being [18]. For instance, for robotic manipulators, it is highly demanded that their hard moves change to soft and flexible motions to suit with their relevant operations. Owing to these, investigating and understanding the human motion mechanisms driven by muscles are essential. In addition, according to Hogan [20], muscles should not be considered force generators since they also play the role of “mechanical impedance regulator” in the movement of body organs. A force generator only moves the organs while an impedance regulator determines the interactions of organs and environment. Moreover, researchers show interests in investigating the ability to execute variable impedance in robotic manipulators. Consequently, various methods have been proposed for VIC which could be generally classified into two main categories, methods based on advanced control including robust impedance control, hybrid impedance control, and adaptive impedance control. Besides, methods were founded on learning and expert systems, in which some examples could be found in the references of [18]. In recent years, the combination of different methods has been found to provide

a more efficient control [21, 22]. Artificial neural network (ANN) and fuzzy inference systems (FISs) are among the most powerful strategies of learning-base methods and are commonly applied in adaptive control of dynamic systems and are able to successfully eliminate or calculate structural and parametric uncertainties of the systems, the effect of external noise, model uncertainties, variable environmental conditions, and other non-ideal factors [23, 24]. Besides, both tools can be used in different steps of one control plan. For example, they can estimate reverse dynamic of robot or determine the model impedance or regulate the parameters of a certain impedance model [25]. Additionally, more advanced methods such as reinforcement learning (RL), interval type II fuzzy system, or a combination of neural and fuzzy methods such as adaptive neural fuzzy inference system (ANFIS) have also been studied and applied in controlling impedance control [26, 27], which are discussed in detail in the “Research background” section.

By reviewing previous studies on VIC in HRI, it is observed that almost all developed models used linear structure. However, the dynamics of a robotic manipulator is strongly nonlinear and the dynamics of human muscles due to generating force and adjusting desired impedance is nonlinear. This shows that they can be more precisely modeled with viscoelastic models with fractional-order (FO) derivatives [28]. Besides, the FO derivatives were found to be more significant than linear methods in controlling the motions of robotic manipulators [29, 30]. Converting the impedance model to fractional helps in resolving the limitations of traditional linear models which are generally applied to enter new modeling spaces, which are less studied in the realm of HRI [31]. However, these results were obtained by implementing simulation model. Also, since the adaptation of muscle dynamics in force generation using viscoelastic models with fractional derivatives has been proved [28, 32, 33], it seems reasonable to transfer the dynamic behavior to HRI which provides further purpose for the application of a model with FO derivatives for impedance control.

Another question arises from environmental uncertainties such as various musculoskeletal dynamics of human users and variations of this dynamics due to various factors such as fatigue, various working conditions and tasks, and the approach of implementing the impedance model to these conditions. To solve this problem, a fuzzy system–based method is a suitable option since it does not require complicated mathematical models and can achieve suitable control performance based on qualitative descriptions represented by linguistic variables [34, 35]. For instance, in linear proportional-integral-derivative (PID) controllers, it is essential to perform at least two numerical derivative signal error calculations which contains errors [36]. Meanwhile, in fuzzy control method, this controller can be executed in a simpler in case of the designing the fuzzy system correctly [37].

Similarly for other controllers, for example, when designing a PID control, implementing the fuzzy logic helps in determining the uncertainty of the system and to define the optimum efficiency coefficients [38]. Furthermore, in order of designing flexible fuzzy system, which can be used in smart applications under variable environmental conditions. It is essential to implement learning methods along with the fuzzy systems to adaptively determine and adjust sections such as relationship functions of input and output variables and/or fuzzy laws based on the data obtained from environment [39]. Besides, other methods become very common in recent years. Application of type II fuzzy system is an interval which considers membership functions with an uncertainty margin to be suitable to improve and support the uncertainties existing in working conditions. Especially, this type of fuzzy system along with fuzzy derivative models is among the key innovations of recent works, and various control approaches have been developed based on these methods [40–46].

In this context, relevant previous research works are investigated. The objective of this study is to investigate recent progress HRI/HRC, with specific attention on VIC approaches for accomplishing smooth, flexible, and adaptive manipulator motion, where the fundamental approaches of VIC are addressed. Identifying that desirable impedance depends of varying tasks (the optimum impedance varies crossway different tasks), the expected impedance in the end effector or robotic grip is parametric, and for further flexibility, it is presented based on a model described by derivative fraction. The fractional order (FO) model is highlighted for the purpose of flexibly parametrize interaction dynamics. Moreover, to handle uncertainty and nonlinear dynamics, fuzzy type and type II fuzzy logic were vastly adopted and addressed. Besides, for tuning the parameters of VIC in real time, hybrid approaches that integrate learning-based approaches parallel with FO and fuzzy logic and proven to be as powerful are also addressed. The current work includes the main concept of research which are illustrated in Fig. 3.

2 Human–robot interaction (HRI) and collaboration (HRC)

HRI is the main concern of this research and with developing new methods to enhance control performance in such applications. Therefore, in the first part of literature review, the recently published research works related to HRI were addressed. There are three different levels of HRI and relations to various interface types as shown in Fig. 4 [47]. These three different levels are defined as illustrated in Fig. 5 (a) and were defined and classified earlier [48]. Else, further classification of HRI based on communication typology is illustrated in Fig. 5 (b) [49]. However, to

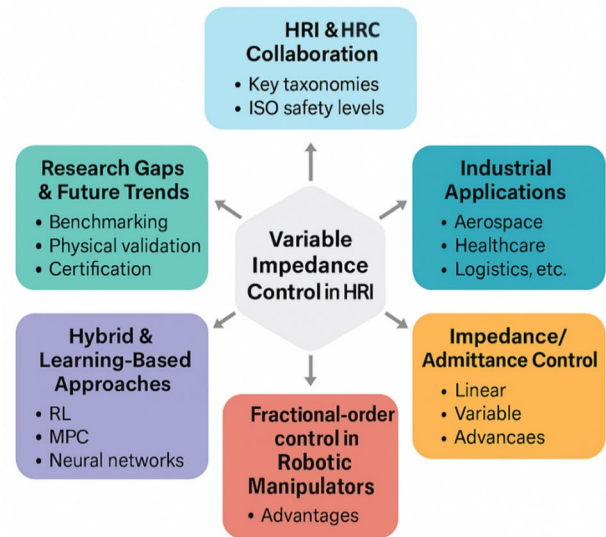


Fig. 3 The main concepts highlighted in research

develop and generate a truly human-centric design, there is still an essential need to evaluate safety, trust, ergonomics, and adaptability within HRI/HRC system [50].

Farajtabar and Charbonneau [51] showed that the physical interaction with humans (pHIR) in HRI systems is significantly boosting with the vast increasing of robotics implementation. The authors attempt includes aspects of perception, planning, and control, and enhancing safety and reliability by the HRC. Besides, they addressed the necessity of worker intention awareness to ensure safety. However, Semeraro et al. [52] studied HRI and the implementations of machine learning (ML) techniques. The term “cobot” for HRC systems was introduced and mentioned that the aim of developed cobots was to increase the productivity and decrease physical burden and mental pressure of workers while working. They considered HRI from two physical and perceptual aspects and defined HRC as “cooperation of humans and robots on the same task in the same place at the same time.” In this paper, tasks of cobots were classified into collaborative assembly, object handling, object handover and collaborative manufacturing and provided separate references for each type. Besides, ML methods in all three types of non-advised, advisor-assisted, and RL methods were discussed, and sensory measurements have been classified into different types including visual, muscle activity, accelerometry, and hybrid. Furthermore, Sharkawy et al. [19] reviewed the implementations HRI and investigated analytical aspects of variable admittance control (VAC) and collision safety. The comparison of VIC and VAC in soft manipulator control methods is presented in Fig. 6. As an example, of HRC in high hazard environment, recently, Tugal et al. [53] investigated the

Fig. 4 Framework to classify HRI levels of interaction and relations to various interface types [47]

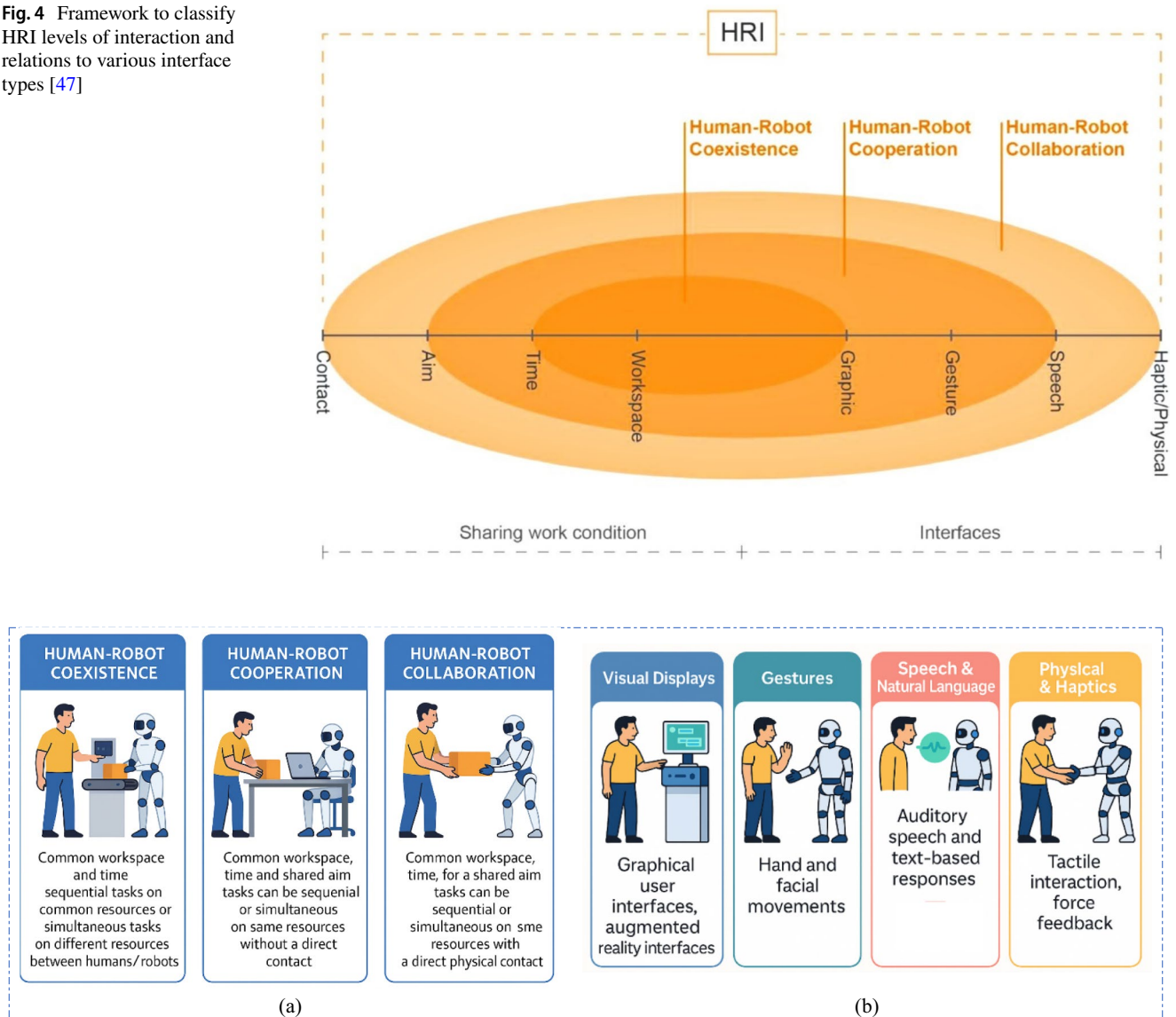


Fig. 5 **a** HRI classifications, this schematic was generated based on the information from [48]; **b** classification of HRI based on communication typology this schematic was generated based on the information from [49]

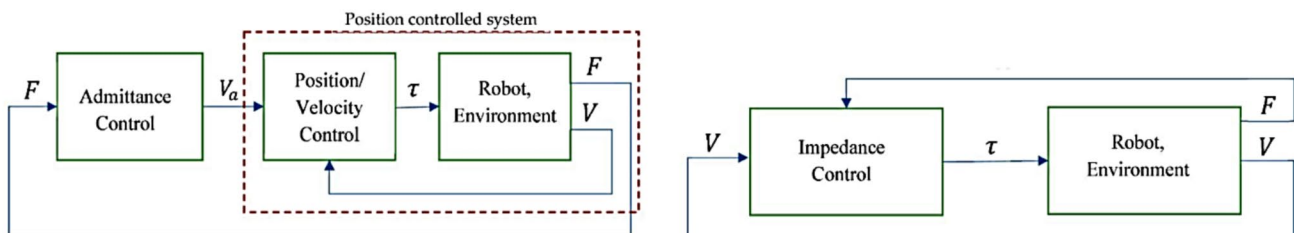


Fig. 6 Admittance control vs. impedance control [19]

performance of cooperation between human worker and remote robots in bilateral teleoperation systems as shown in Fig. 7 crossways various telerobotic tasks. Where the

stationary gaze entropy (SGE) and gaze transition entropy (GTE) were used to capture gaze shifts randomness analyzing dwell-time complexity.

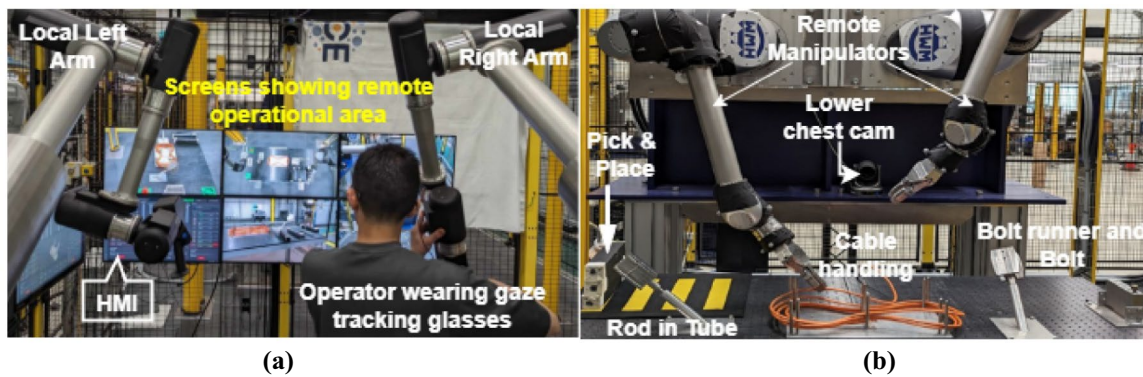
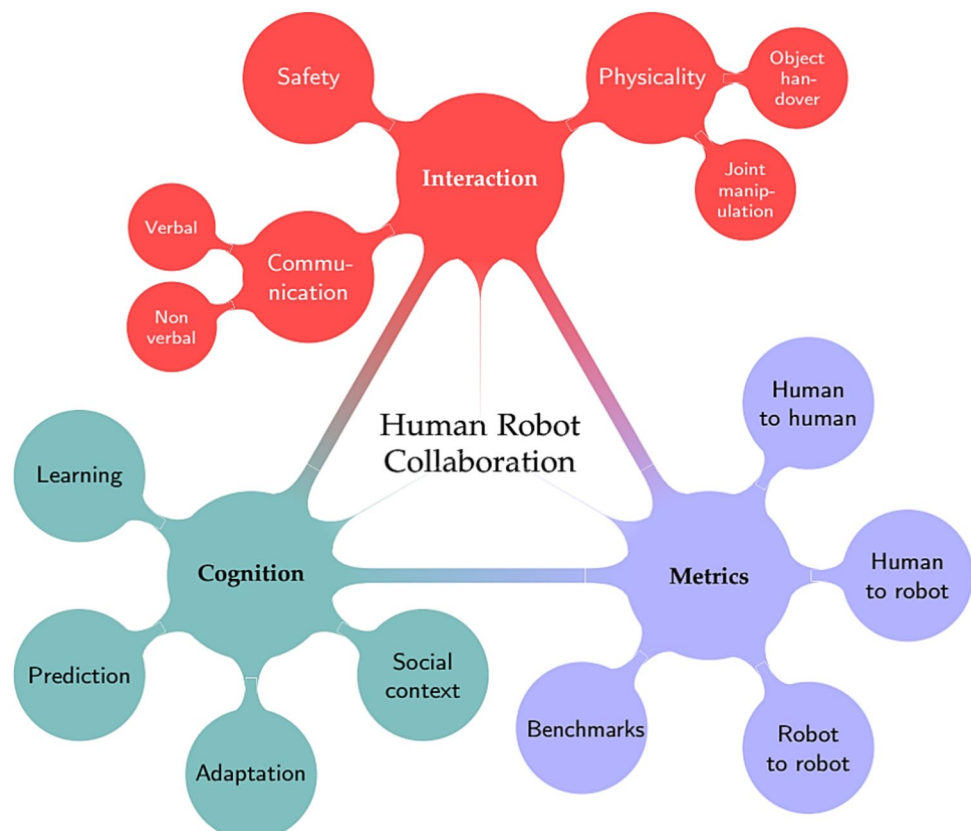


Fig. 7 Bilateral telerobotic system with telbot dual-hand, (a) local side and (b) the remote side, featuring several tasks utilized in the study [53]

While in HRC systems, a high sensitivity must be insured by robots to any delicate human motion, parallel with minimizing safe interaction during the collaboration, methods as infinite impulse response filter revealed high performance in detection of low–high frequency range based on human motion compared with Fourier transformation [54]. Moreover, Castro et al. [55] explored the utilization of HRC in industrial field, object handover, and learning and success criteria, in which a detailed discussion on the definition and distinguishing of HRI and HRC was presented and briefly defined “interaction” of one task on another and “collaboration” with others.

The interaction as a part of HRC has been considered with providing perception and indices for successful execution of HRI, as presented in Fig. 8. Besides, the safety in work environment was also discussed, and levels of cobot safety based on ISO10218-1 and ISO10218-2 standards (which are accumulated to the recent ISO 10218–1:2025 [56] and ISO 10218–2:2025 [57]) were defined, in which at least one level should be applied, as demonstrated in Fig. 9: a) safety-rated monitored stop, b) robot movement imitation by hand guiding, c) speed and separation monitoring (depending on human proximity, robot reduces speed), and d) power and force

Fig. 8 Relationship map of HRC and HRI and subsets of interaction, perception and indexes [55]



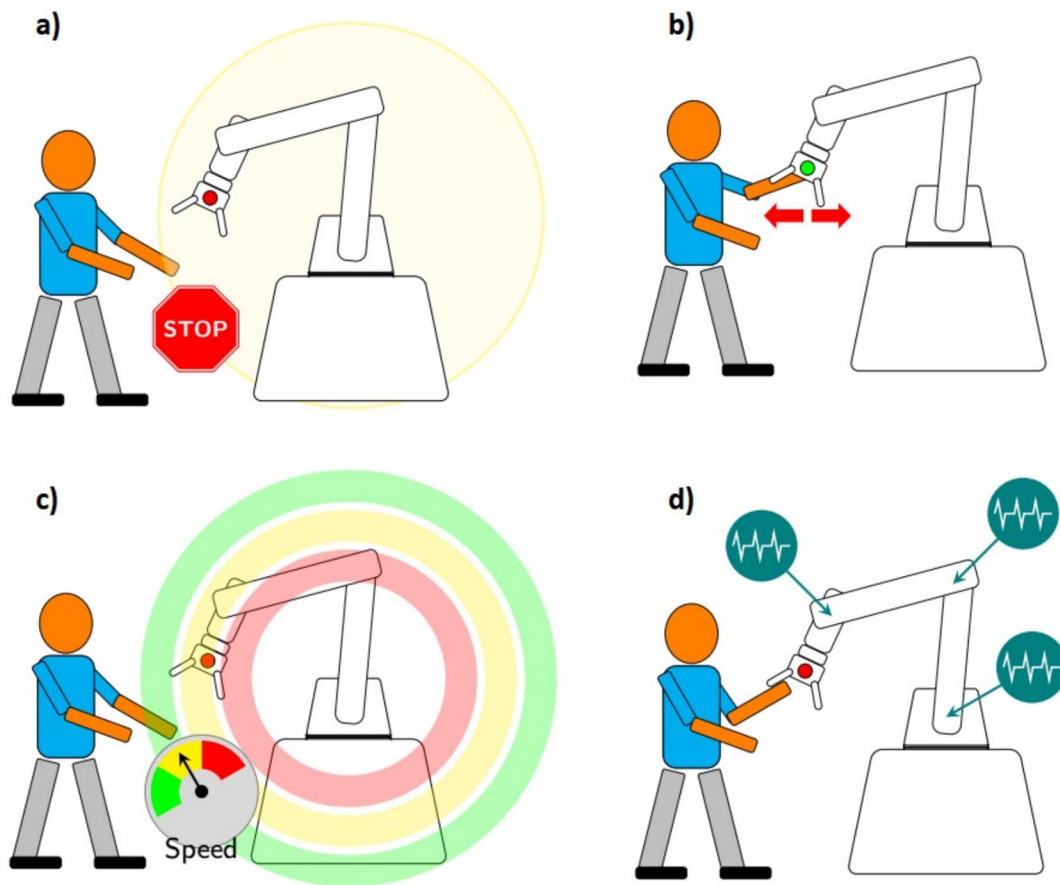


Fig. 9 Safety levels in coots based on ISO20118-1 and ISO10218-2 standards [55]

limiting. Besides, the HRC was classified into two categories of “goal-oriented” and “person-oriented”; such factors as success rate, interactive force, timing, and joint effort are in the first category and factors such as fluency, satisfaction, comfort, trust on robot, and human-like motions are classified in the second category. However, in the context of biomechanical risk in HRC, Silveti et al. [58] used statistical data analysis to prove that the integration of cobots assists in minimizing the biomechanical risk for labors, mainly right wrist, shoulders, and trunk injuries.

Furthermore, Mukherjee et al. [59] investigated the robotic learning strategies of HRC. It was found that the number of publications in the period 1996 to 2020 on HRC was significantly high and includes four main core classifications, which are HRC, HRC industry, HRC ML, and HRC industry ML, as illustrated in Fig. 10. Besides, the HRI from 0 to 5 has been classified into two extremes. In the first extreme, the robot was located in a protected chamber and performs predetermined tasks, while in the second extreme, collaborating between worker and robots at the same task in the same surrounding, as illustrated in Fig. 11. These correlation levels were L0 fully programmed, in which operation is terminated when a human is in the work place with the robot. Besides, in L1 coexistence,

in which in working environment, robot, and human are not totally separated and in case if human is presented in work environment to collaborate with the robot, the robot does not completely terminate its operation as reducing robot speed. In L2 assistance, the robot supports humans to perform heavy duties such as lifting or carrying heavy objects. L3 cooperation is at a higher level compared to L2 assistance in which robots cooperate with humans in a consecutive way although no physical contact takes place between human and robot. In L4 collaboration, humans and robots do a task in the same station at the same time with the same goal. At this level of collaboration, no protected space exists and robots and physical contact can be performed. However, robots learn tasks through several methods such as observation or ML. Similarly, instead of physical contact, other communication forms such as voice, visual contact, pointing, or body language can also be used. Additionally, in L5 fully autonomous, the robot generalizes the tasks learned by ML to perform several tasks without direct human intervention, which can impose a certain style in doing tasks.

In another study, Rodríguez-Guerra et al. [4] discussed the increasing of HRC utilization in industrial working environments; challenges and suggested solutions were also discussed. In that study the HRC was considered a difficult

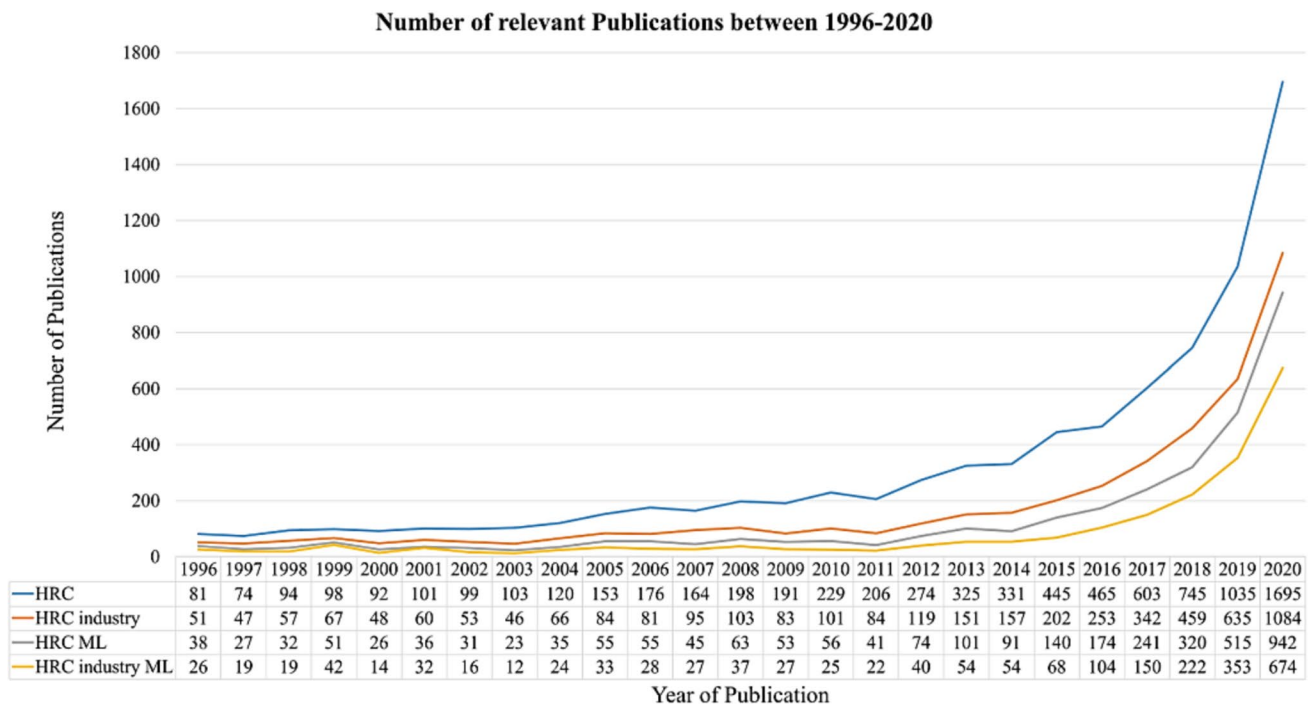
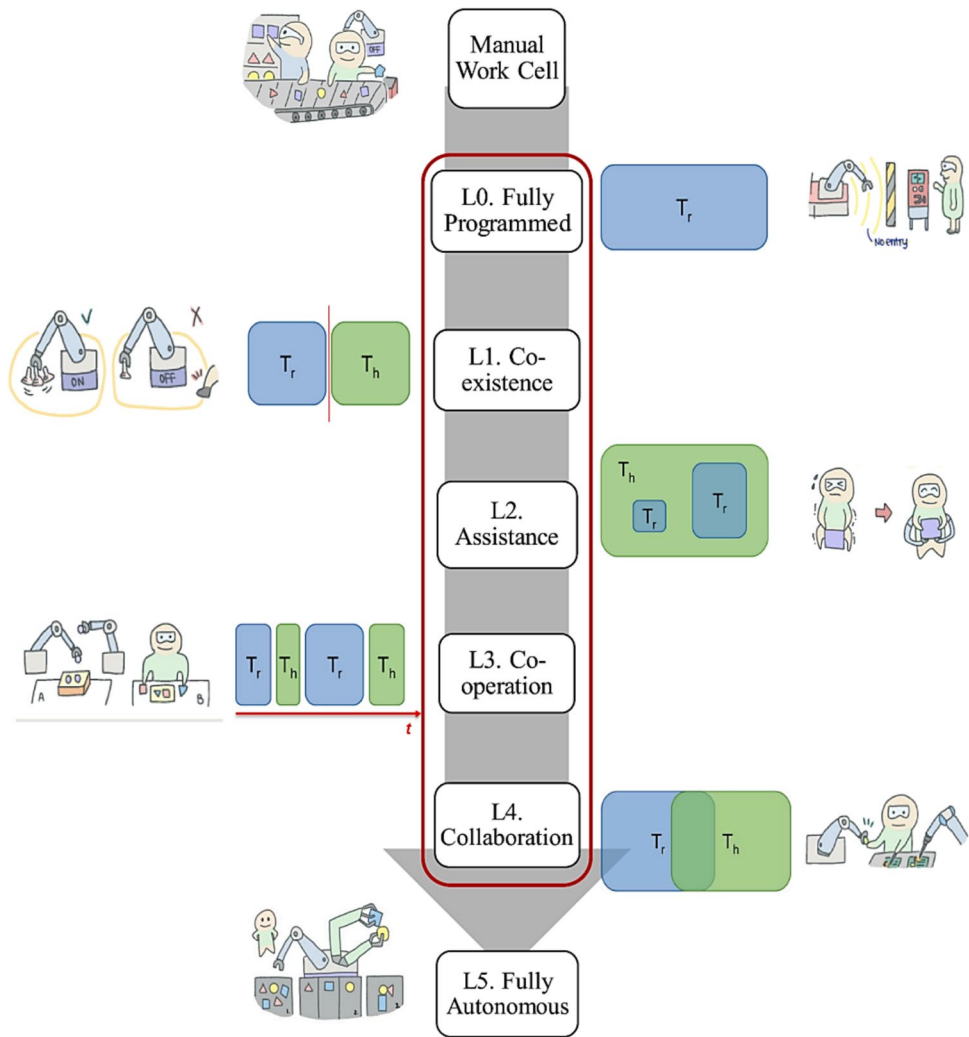
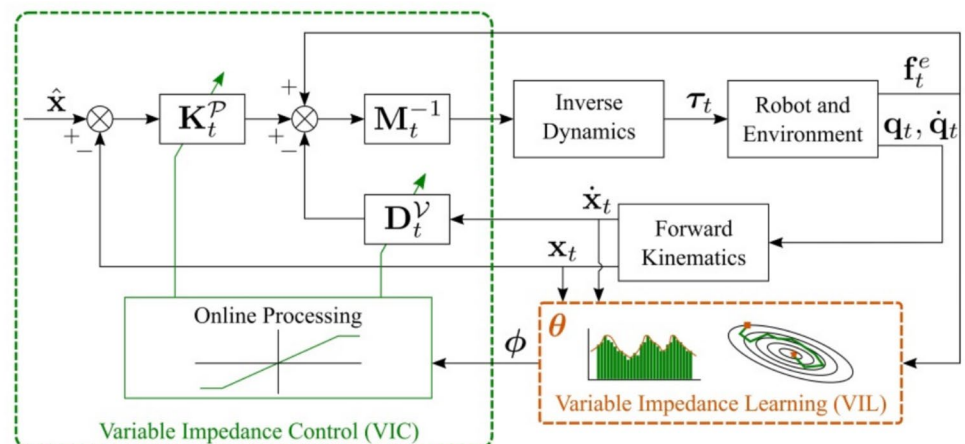


Fig. 10 Number of publications in the period of 1996 to 2020 on HRC [59]

issue since an unpredictable behavior of a worker can result in dangerous and harmful physical contact. Therefore, to address this issue and respond to unpredictable behaviors with uncertainties, a classification of four categories was accomplished which are operation, industrial processing, task design, and working cell design. Among all these categories, the task design category has been studied more than other categories. Besides, a task can be simple movement between two points, grabbing a part, or passing through collision through obstacles. Moreover, operation is another topic and covers a series of functions which are assumed by performing various tasks to accomplish a certain goal. In the classification presented in this paper, operations are divided into two levels of “physical interaction” and “perceptive interaction.” After defining HRC challenge levels, solutions are addressed. Furthermore, Gervasi et al. [60] suggested a conceptual framework for HRC, in which the collaboration was defined as a subset of interaction and the HRC was defined as a type of HRI with the aim to combine the humans’ skills with robots’ skills. Besides, the ISO/TS15066:2016 standard was considered in consequence; the collaboration was considered a type of physical interaction with human in a common environment in which the robot has at least one degree of autonomy (or smartness) and human safety is guaranteed in this interaction. In that conceptual framework, the up-to-bottom method was used to present the issue of HRC to investigate the hidden aspects of this relationship. However, eight hidden aspects

were considered including human factors (ergonomics), autonomy, team organization, information exchange, work, adaptability and education, morality, and cyber safety for HRC and described each of them by defining different levels for each item. Interestingly, AboZaid et al. [61] accomplished a comprehensive review that specifically focused on implementations of soft robots in HRI and their supreme characterizations from different perspectives, such as fabrication process made of stretchy and soft materials with nearly unlimited degrees of freedom (DOF) a significant flexibility. Besides, several real-life examples were illustrated to present the superior ability of soft robots in analyzing living organisms’ motion and provide safe environment with humans. Moreover, Sharifi et al. [62] reviewed VIC and learning strategies in HRI. Similarly, a further study on the same topic was presented by Abu-Dakka and Savarano [63]. In both studies, the method of “learning from demonstrations,” which is a simple and powerful learning tool, was considered for the processing of data collected from several experimental observations based on which the robots’ imitated humans. When desired impedance learning is considered for a task, in terms of mathematics, impedance learning means finding a mapping (overly nonlinear) to map the observed data to assumed coefficients in impedance model, for example, assuming hardness and oscillation coefficients in variable impedance model, finding mathematical mappings for φ^k and φ^D from the measured data of position, speed, and external force exerted on the end effector of the

Fig. 12 Schematic diagram of variable impedance learning [63]


$$\begin{aligned}\hat{\mathcal{K}}_t &= \varphi^K(\mathbf{x}_t, \mathbf{x}_{t'}, \mathbf{f}_t^e, \theta^K) \\ \hat{\mathcal{D}}_t &= \varphi^D(\mathbf{x}_t, \mathbf{x}_{t'}, \mathbf{f}_t^e, \theta^D)\end{aligned}\tag{1}$$


Additionally, Keemink et al. [64] reviewed impedance control in HRI, in which humans are introduced as a special case in the environment in case physical interaction with robots is performed. Besides, the admittance control was introduced as the counterpart or opposite of impedance control and advised important points in applying each of them: admittance control can present a virtual hard surface significantly, although it has challenges in delivering low inertia. Consequently, complexity may appear when facing a real hard surface (restricted movement). In contrast, impedance control can deliver low inertia significantly but has limitations in delivering virtual hard surface. Therefore, in a dynamic interaction with low inertia (free movement), complexity may appear. The admittance control was used in a general manner for all relationships of force to desired motion and mentioned that admittance control can simultaneously follow force and desired displacement, as illustrated in Fig. 13. However, enhancing HRI for supreme instinctive requires recognizing and applying subtle human signals, like micro-movements, to convey robots' intentions [65]. Research in HRI gains from incorporating effective strategies from psychology, including control and priming techniques, to develop strong and widely applicable studies [66].

3 Industrial applications of cobots system

Recent developments in cobot systems go beyond the conventional automation systems, and their industrial applications are increasing drastically. The cobots are designed based on essential considerations such as safety interactions with labors in various working environments during several working operations. Besides, the ability of cobots in integrating several modern control strategies such as fuzzy logic, adaptive impedance control, RL and admittance control. The integration of such strategies permits robots to regulate variable payload, erratic environments, as well as human inputs in actual time. To provide an insight of various applications of cobots in industries, this section addresses some industrial application examples from literature in different areas such as aerospace, automotive, healthcare, logistics, and electronics. However, the term industrial robot must meet the ISO

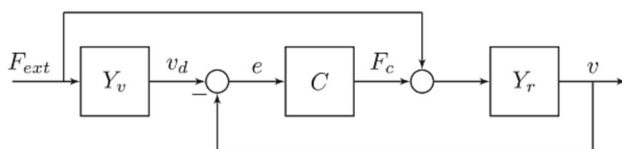


Fig. 13 Structure of admittance control in which external forces deliver desired movement by passing through a virtual dynamism (desired admittance model) and the controller should apply it to the robot by calculating force/torque. Also, this external force is directly applied to robot dynamism [64]

8373:2012 standard [67] (which is accumulated to the recent ISO 8373:2021 standard) [68]. However, the key main applications of cobots that are common in several manufacturing industry sector are illustrated in Fig. 14, while Fig. 15 shows the overall workflow of any industrial process between human labor and cobots. Various examples are available in literature about the interaction between human and robots in cobots system. For instance, an extreme transformation was accomplished in aerospace operation through integrating the cobots due to the achieved improvement during assembly, inspection, safety through maintenance, repeatability, and precision [69]. Futterlieb et al. [70] presented one of the most significant air-cobot projects, where the authors developed a collaborative mobile robot to gain a human–robot inspection of a fuselage across maintenance operations before takeoff. They also address essential design consideration for applications of cobots in aerospace, as multi-sensor fusion, SLAM-based navigation, severe integration within regular maintenance, and dynamic problem avoidance. Raviola et al. [71] developed an automated smart sequence system to extract various health parameters of unit under test (UUT) and highlighted the utilization of cobots equipped with a vision scheme and customized tools for both maintenance and health parameter extraction on unserviceable servo-actuators. Meanwhile, in logistic perspectives, Puttero [72] presented a comprehensive literature where practical case studies were addressed to understand the integration of cobots in quality control by highlighting their restriction in implementations. Besides, the analysis of case studies came to conclude five main criteria for utilizing cobots in quality

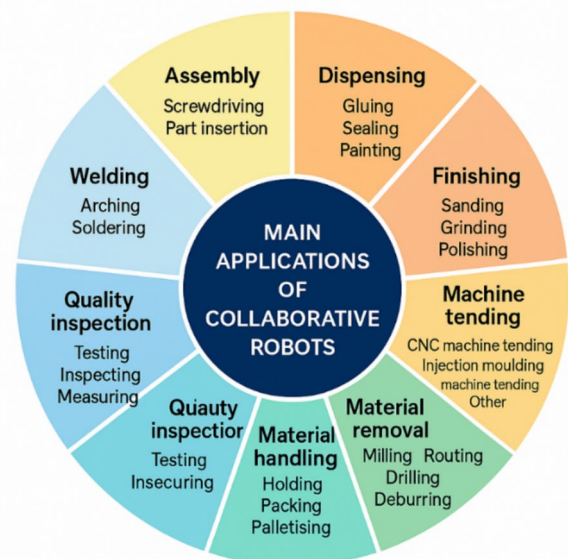


Fig. 14 Key main application of cobots

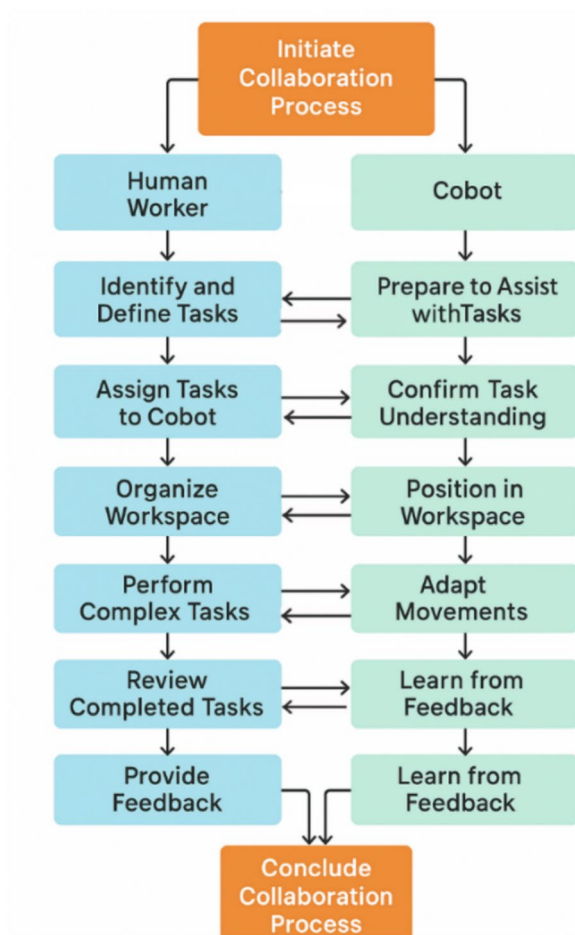


Fig. 15 The workflow process between human labor and cobots

control, which are quality control paradigm and inspection type, the insights of visual inspection, trust and safety in collaborative environment, human factor versus cobots efficiency, and growing economy. Raji et al. conducted a review study that focused on applications of cobots in rehabilitation. The authors highlighted the programming and safety features of commercial cobots. Besides, various types of used systems were addressed, such as ABB YuMi, KUKA LBR, and UR5e. Further studies and industrial applications are presented in Table 1. Besides, Figs. 16 and 17 illustrate a visual example of these applications.

4 Impedance/admittance control

Admittance control modifies the robot's movement based on external forces, making it ideal for activities with necessary compliant behavior [102]. Besides, the attributes of the human operator and the differences in physical strength and skill level when interacting with a robot are also essential to be considered [103]. Unified interaction control blends impedance and admittance control to enhance performance according to the characteristics of the robot and its surroundings [104]. Therefore, Huang and Chen [105] focused on soft movement of manipulators. The end effector to be in direct contact to a geometric surface such that an interactive force generates between them while doing the task was considered. Adopting a linear impedance model with coefficients of mass, oscillation, and hardness with constant values and analytically evaluated the possibility of reaching this desired impedance in the manipulator. Besides, the concept

Table 1 Cobots applications in various industrial sectors

Application	Key implementations	Refs
Manufacturing	Teleoperation Wearhouse Polishing	[73–82]
Healthcare	Invasive surgery Monitoring Rehabilitation Pharmaceutical Patient care	[83–89]
Electronics	Defecation smart care robot	[90]
Automotive	Mobile welding robot Human-centric automation	[79, 81, 82, 88, 91, 92]
Food and beverage	Packaging and palletizing	[88]
Aerospace	Diagnostic Automatic maintenance	[81, 93, 94]
Various environments	Dynamic environment	[95]
Logistics	Workforce transition	[79, 82]
Agriculture	Safety perception Packing	[79, 96]
Constructions	Construction process	[97]

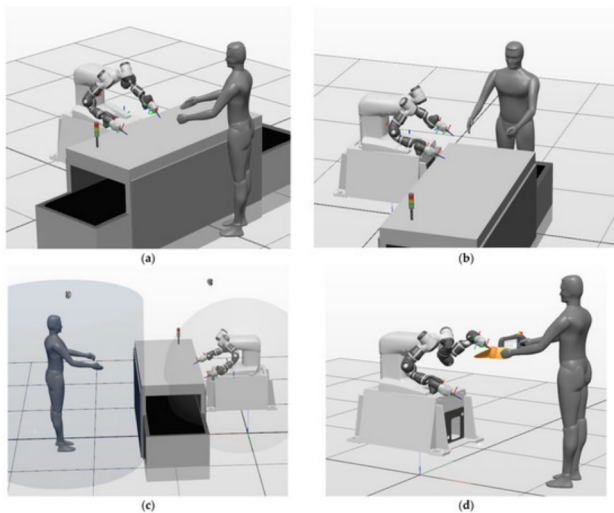


Fig. 16 Cobots in various stages: **a** monitoring station with considering safety, **(b)** direction and hand guiding, **(c)** speed and separation monitoring, and **(d)** torque and force [80]

of VIC and developed two methods for that based on position feedback and force feedback was presented. Although the developed theory did not directly consider the interaction with human, its principles were applicable to HRI. Moreover, Song et al. [18] performed a detailed review focusing on robotic manipulator impedance controllers', in which the manipulator control methods have been classified into four main groups of position control, impedance control, force control, and hybrid position/force control. Position control is applied in cases where non-contact movement or free-end effector motions are considered [106]. In contrast, force control is applied where a contact force is expected between end effector and the surrounded atmosphere. In position control, any deviation from the desired rout is strictly avoided; therefore, motion control increased with impedance. However, in force control, end effector displacement is not restricted and the only goal is preserving a certain contact force; therefore, motion control is performed with low impedance [107]. Accordingly, position and force control are considered two impedance control extremes. Nonetheless, in hybrid control, motion space of end effector is divided into two free and restricted subspaces and in each subspace, relevant control is applied [108]. For example, position control is performed in free subspace and force control is performed in restricted subspace. Hybrid control requires accurate knowledge on the geometry and specifications of work space [109]. However, in impedance control, position and force are controlled simultaneously and are not separated. In fact, position control is performed in such a way to produce the desired amount of interactive force in the end effector based on a predetermined model called impedance model. Impedance model is generally a linear model described as [18]

$$M(\ddot{x} - \ddot{x}_d) + B(\dot{x} - \dot{x}_d) + K(x - x_d) = F(t) - F_r(t) \quad (2)$$

where x_d is desired rout, F_r is desired force, x is actual rout, and F is the actual force of end effector.

Admittance control is, in fact, the binary of impedance control, and while impedance model generates a causative relation between motion (as the cause or input) and force (as the effect or output), admittance control creates a causative relation between force (as the cause or input) and motion (as the effect or output) [18, 64]. Furthermore, two methods are available for performing impedance control, establishment of hardware and establishment of software [110]. Well-designed hardware parts are used to obtain soft dynamic response from the robot [111]. For example, a spring or adjustable oscillator in joint structures of robotic manipulator, software establishment is applied. This method is applied in designing smart artificial organs, robotic rehabilitation, moving-aid devices with help-when-needed strategy, and artificial pneumatic muscles [112]. However, in the software method, a controller applies a desired impedance, in other words, the expected dynamic softness in robotic response through controllable drivers which manage system energy [113]. The software method is executed upon receiving position feedback and the force exerted on the end effector [114]. Therefore, a force/torque sensor is necessary for executing impedance control [115]. Moreover, in the software method, two control loops, called external and internal loops, exist. In external loop, desired motion route for the end effector is determined based on impedance model in Eq. (2) to obtain the desired dynamic softness at interaction point of robot and environment by moving (tracking) on that route. Internal loop is only responsible for executing the tracking of the route determined by the external loop, which is labeled as position servo system [116].

4.1 Variable impedance/admittance control

VAC is employed to modify the robot's reaction to external forces by adjusting the damping and stiffness parameters according to the interaction dynamics, which make the VAC ideal for environments with varying stiffness, such as during collaboration with human arms in HRC [117]. VIC and VAC enhance HRC by improving performance and stability in complex processes and unstructured environments [118]. Besides, VIC and VAC facilitate robots exhibiting variable compliant behaviors, minimizing contact force, and safeguarding against leg damage when faced with static or dynamic disturbances. Furthermore, initial interests in considering human robotic manipulator collaboration and applied VIC method was recorded by Ikeura et al. [119]. In collaboration, the robot helps human in lifting the object, and a linear impedance model with constant mass and variable oscillator coefficients was considered and applied

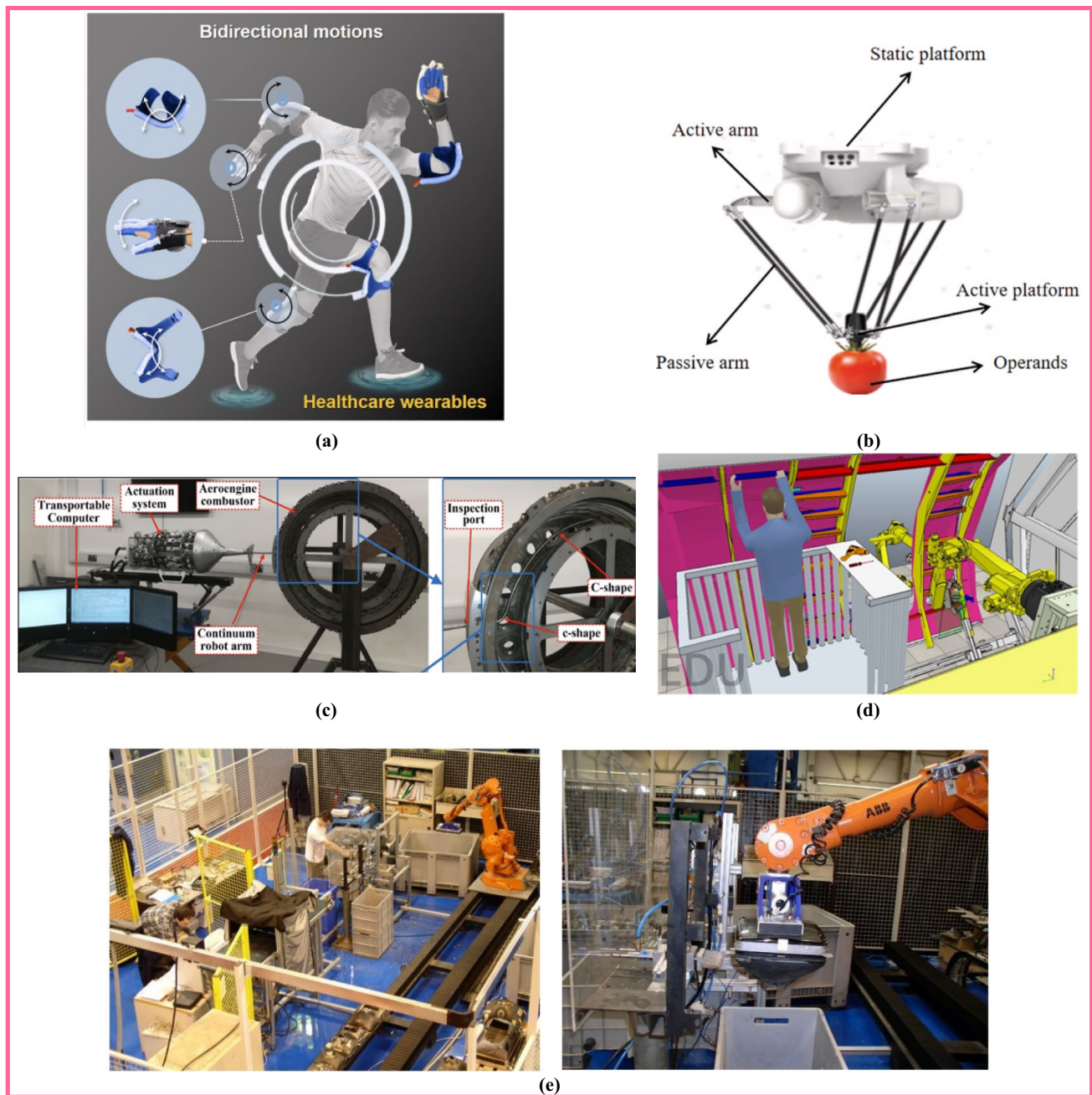


Fig. 17 Examples of cobots applications in various industries: **a** healthcare [98], **(b)** agriculture [96], **(c, d)** aerospace [99, 100], and **e** recyclability and high risk task [101]

oscillation coefficient in an optimal and analytical manner based on minimizing oscillation coefficient and its derivative in a certain time interval were determined.

In the recent decade, several studies have been performed on variable impedance/admittance. Some studies are reviewed in this section, and other studies are presented in the next sections since they specifically used FO derivatives, fuzzy system, or learning methods. However, Ferraguti et al. [120] introduced a new method called reservoir-based

technique for VIC, in which the inactivity of impedance control method was highlighted followed with presenting in which variable impedance model, hardness coefficient is chosen to be invariable with time. Inactivity criteria are desecrated which can result in instability of interactive force and not following the desired route by the robot. Then, to address the problem of one variable, a new status with adaptive dynamic along with their linear impedance model to act in a specific manner as a storage reservoir was

addressed to inject excessive energy into the system to use it for changing impedance hardness. Besides, that addition of such subsystems to the overall impedance control system can prevent instability of interactive force and desired tracking. However, Ghanbarzadeh and Njafi [121] developed a new VIC model for HRI system based on ISO/TS 15066 standards to adjust damping parameters and stiffness dynamically. The model achieved 78% and 18% improvements in speed on three-joint and Panda robot respectively. Furthermore, Kronander et al. [122] studied the stability in VIC. Compared with the work presented by Faraghti et al. [120], a developed method was presented as independent from status variable and the concept of energy reservoir. In addition, both oscillation and impedance hardness coefficients were considered variables and proved stability based on Liapunov method. Moreover, Duan et al. [123] developed adaptive VIC to track contact force in environments with uncertainties (specifically uncertain geometry and hardness). Such approach permits the robot end-effector to apply a variable (i.e., dynamic) desired force to the environment during motion, which the authors defined as dynamic force tracking. However, the authors claimed that

this was an innovation since for the first time, environment was considered to have uncertain conditions. They verified the correctness of their developed method by proving stability based on Roth method, simulation, and experimental tests as seen in Fig. 18. Interestingly, to enhance compliance HRI, a sensor-less VIC has been proposed by Liu et al. [124]. That approach leverages end-effector velocity and a compensation model to infer interaction forces and adjusted the impedance parameters dynamically without integrating physical force sensors. Besides, Lyapunov theorem was utilized to define the stability, while the simulation outcomes revealed improved compliance and velocity tracking. That study proposed a practical sensor-free alternative within VIC field. Furthermore, Shen et al. [125] investigated the industrial operation such as sanding and finishing with relatively high contact force using VIC, where a fuzzy-based adaptive impedance capable of controlling robotic manipulator was suggested, so that it can sand and polish parts with different materials with deskside. Besides, variable parameters in the impedance model were considered to be oscillation and hardness. These parameters were adjusted at optimum values using repeating minimum recursive squares method. In

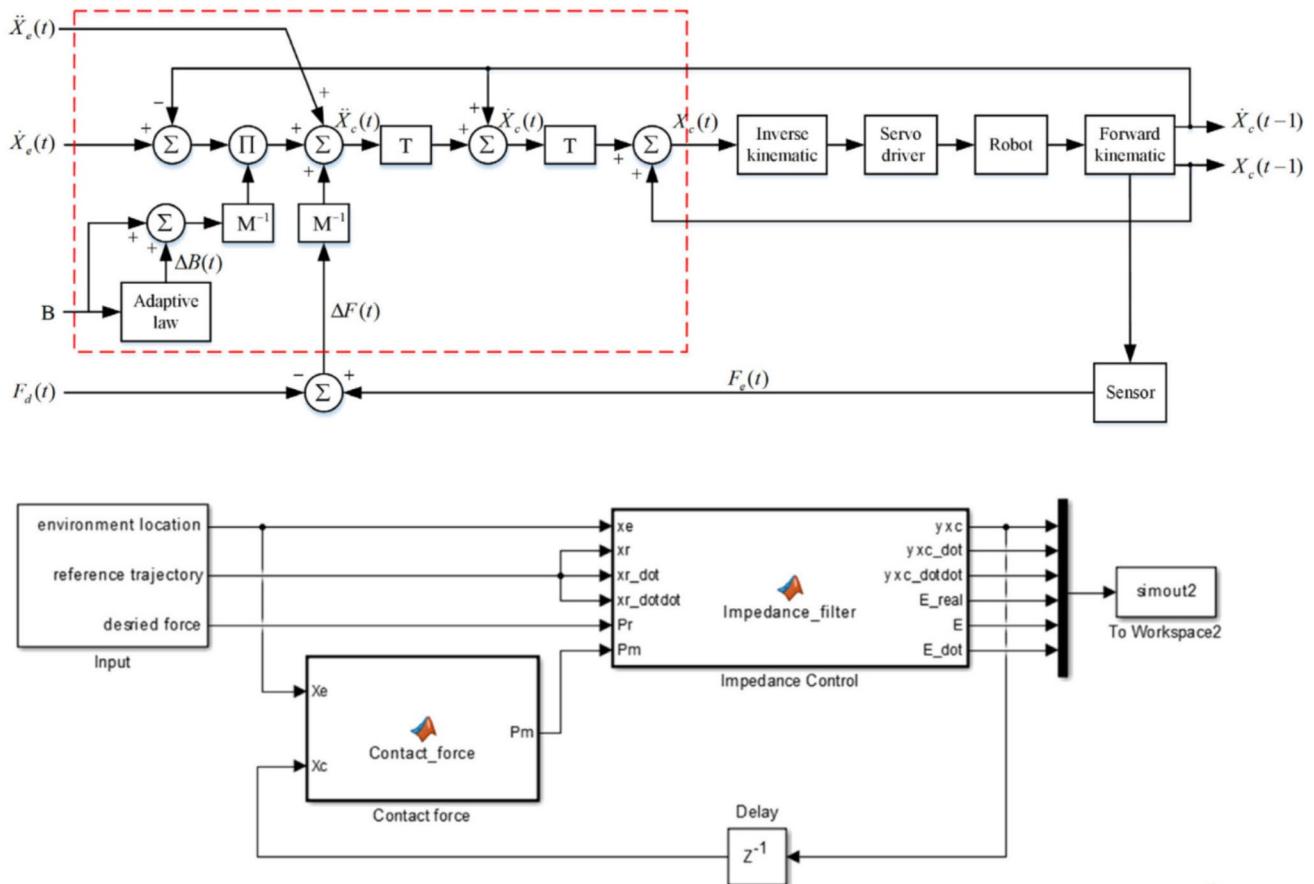


Fig. 18 Diagrams of control and its simulation in SIMULINK MATLAB software [123]

this control design, the fuzzy system was applied to suggest an appropriate contact force without primary knowledge on the environment. On this basis, by estimating environmental parameters by fuzzy system, impedance model parameters were adjusted. The implementation of VIC often leads to the introduction of energy-injecting components, which may affect the stability and passivity of a robotic system. In contrast, VAC encounters difficulties differentiating between divergent forces independent of errors and those based on mistakes [118].

4.2 Limitations and developments of linear impedance models

4.2.1 Limitations of linear impedance models

Linear impedance models are considered the basis of impedance/admittance control in human-robot interaction due to their unique damping parameters and constant stiffness characterizations, and their simplicity and ease of implementation. However, these models fundamentally assume linear force-motion relations that are unsuitable to capture physiological inconsistency of the human musculoskeletal system and the complexity in real environments. He et al. [126] confirmed that a nonlinear, hysteretic characteristics were exhibited by human joint impedance. Such features cannot be captured by linear models; this limitation in linear models have the potential to compromise the stability of interaction and the accuracy of force estimation. Besides, the

linear impedance models are not ideal choice in system that might have unpredictable interaction or at high-frequency range, yielding to trail errors and decreased compliance, mainly in tasks demanding quick acclimation as collaborative assembly processes. In addition, an experimental evaluation accomplished by Sharkawy and Koustoumpardis [19] revealed that in case of subjecting the linear model to a system with unexpected variation in contact condition, a weak performance and disposed to force overshoots might appear. These observations collectively address the fact that implementing linear parametrization might cause serious challenges in exact estimation of main interaction forces, specifically in changeable HRI systems. Figure 19 illustrates the main limitations of linear impedance models.

4.2.2 Robustness and accuracy of nonlinear impedance models

To overcome the limitations of linear models in HRI, the nonlinear impedance models rise as a significant alternative, mostly owing to their suitability for complex systems, task-dependency force dynamic, and physiology of human bodies. The nonlinear impedance models permit further accurate adaptation by integrating non-constant stiffness and damping features, frequently modified by contextual signals or physical signals. Jiao et al. [127] revealed that the nonlinear model outperforms linear model in aerial robotic task where it accomplished greater tracking fidelity and responsiveness. Similarly, Tiseo et al. [128] proposed a novel model that is

Fig. 19 Key limitations of linear impedance models in HRI and HRC

Key Limitations of Linear Impedance Models in HRI and HRC



able to strongly handle signal noise and contact transitions, representing minor trajectory errors compared with linear model in both experimental and simulation. Furthermore, Zou et al. [129] implemented the nonlinear gain-scheduled model combined with frequency domain passivity criteria. By investigating at high-frequency force inputs or under changeable interaction environment, it was signified that nonlinear models achieved extreme accuracy and robustness compared with linear models. The main capabilities of nonlinear impedance models are demonstrated in Fig. 20. On the other hand, Fig. 21 illustrates the main criteria for robustness and accuracy in modern nonlinear impedance models for HRI and HRC.

4.2.3 Current methods of nonlinear modeling

Modern development in impedance significantly increases the implementations and strategies of nonlinear models such as fuzzy logic, FO dynamic, and learning methods to higher accuracy model in diverse cobots interactions parallel with enhancing the estimation of dominant interaction force. For instance, a subsystem based on adaptive nonlinear impedance controllers has been proposed by Hejrati and Mattila

[130]. The aim of the model was to provide virtual stability and accomplishing wider Z-width during contact-rich tasks in a 7 DOF haptic exoskeleton. Likewise, to estimate humane body motions, Dong et al. [131] used the fuzzy logic with tune impedance feature in real time, confirming uniformly ultimately bounded (UUB) trailing despite model uncertainties. In a complementary technique, Ding et al. [132] implemented the FO impedance controllers to enhance damping representation and robustness. The experimental work was conducted with robotic manipulators, and the obtained results revealed faster settling times, lower overshoot, and supreme absence of disturbance in contrast to integer-order models. Moreover, the interaction of human–robot with non-linear modeling has been enhanced with the advanced RL approaches. For instance, Roveda et al. [25] presented developed impedance model that consist of Q-learning-based model-predictive VIC (Q-LMPVIC). This model has the ability to regulate damping features and Cartesian stiffness by utilizing learned model by neural networks (NN) and stabilized by Lyapunov constrains. Such technique is capable of enhancing the accuracy of force estimation and collaborative control flexibility through real process. Figure 22 presents the current approaches in addressing nonlinear modeling.

Capability	Description
Real-time Adaptability	Learns from user motion and adjusts impedance dynamically
Robustness to Perturbations	Maintains performance under noise, task variation, and uncertainty
Neuro-inspired Precision	Models co-contraction and muscle behavior for stiffness tuning
Hybrid Control Integration	Combines neural nets with sliding/optimal control for better accuracy
Enhanced Signal Decoding	Improves bio-signal interpretation and human-intent tracking

Fig. 20 The main capabilities of nonlinear impedance models

Fig. 21 Key attributes of robustness and accuracy of nonlinear impedance models in HRI/HRC

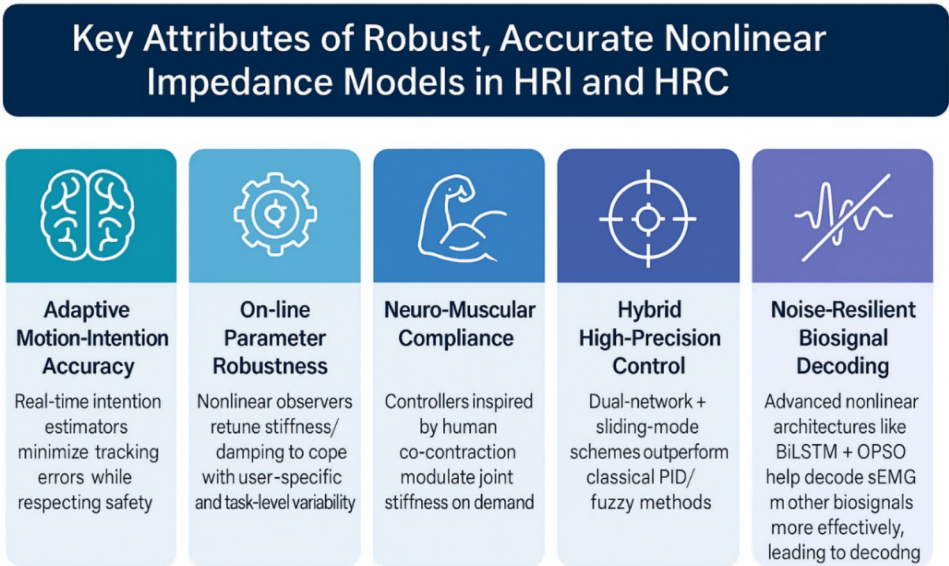
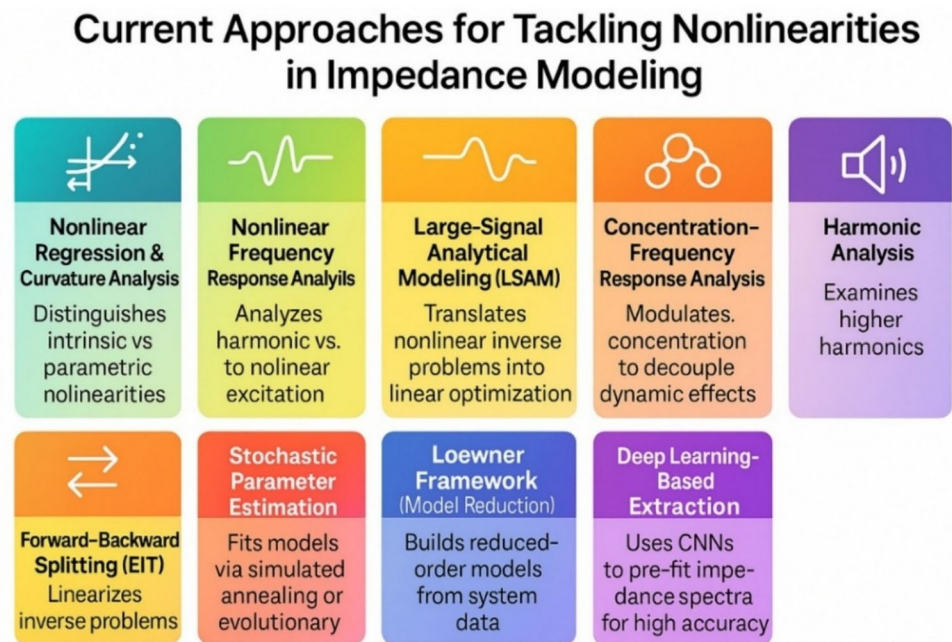


Fig. 22 Current approaches for nonlinear modeling

4.2.4 Trade-offs between model complexity and accuracy

The implementation of nonlinear impedance types has constantly exposed a trade-off between improved interaction force estimation and greater system complexity. For example, for unknown environment, Gu and Huang [133] revealed that implementing the OF PID controllers causes to enhance speed response and the force tracking accuracy recorded (29–60)% higher values compared with adaptive VIC and adaptive hybrid impedance control. Besides these enhancements, a supreme reduction in overshoot has been further noticed. In addition, Babarhamati et al. [134] proved that enhanced force tracking with the absence of virtual energy buffers and enhanced trajectory combined with strong numerical behavior and passive were all achieved by utilizing fractal impedance frameworks. Such features make the controller suitable for applications, demanding robust dynamic interaction at the end effector. Anand et al. [135] developed a new technique known as deep model predictive variable impedance controller (MPVIC) that included learning generalized Cartesian impedance model for a robot manipulator in order to increase the gained information to optimum. Besides, the Franka Emika Panda robot was utilized to determine the efficiency of the developed technique. The obtained outcomes revealed that implementing NN and model predictive control (MPC) optimizer enhances the accuracy and force estimation with eliminated necessity of retrain crossway new task in case of used in multi-task operations. Hammoud et al. [136] considered the risk sensitive control to analyze uncertainty and produce a formal path to involve uncertainty for unidentified contact location. The

investigated model accomplished a remarkable progress in performance during trotting and jumping tasks with virtual quadruped robot. Kang et al. [137] highlighted the challenges between preserving robustness against uncertainty in modeling and accomplishing high accuracy, where also the delicate accurate tuned controllers were addressed. Besides, a transitional filtering strategy was proposed to facilitate this trade-off. Furthermore, to investigate the performance accuracy in a dynamic environment, Xiao et al. [138] used deep learning object detection (SSD) to create simultaneous localization and mapping (SLAM). Meanwhile, for positions based controller was employed through using radial basis function neural network (RBFNN) combined with gradient descent approach by Li et al. [139], where the accuracy of position control shows significant enhancement. Figure 23 illustrates some examples of accuracy and complexity of some complex technique.

5 FO control in robotic manipulators

Incorporating fractional derivatives into sliding mode control (SMC) strategies has demonstrated improved performance in trajectory tracking tasks, even when handled with external disturbances [148]. This method utilizes the distinctive characteristics of fractional calculus to improve control accuracy and resilience. FO admittance control techniques have been suggested to ensure precise robot-environment interaction with minimal overshoot and rapid response. This renders them practical for industrial and medical applications requiring precise force control from robots [149]. Focusing only on FO

Complex Technique	Accuracy	Complexity	Ref.
Nonlinear optimal control	Supreme disturbance rejection and adaptability	Online stochastic optimization	[140]
Lyapunov control	Steady performance in flexible systems	Advanced modeling and stability proofs	[141]
Filtering theory	Balances precision and robustness	Designing filters and iterative tuning	[137]
FO, Fuzzy, RL, NN	Flexible and Robust characteristics tuning in real-world	Rule-tuning, training iterations, fractional calculations	[62, 142–145]
Gaussian Processes	Effective adaptation with small data set	High-dimensional deduction complexity	[146, 147]

Fig. 23 Examples of accuracy and complexity of some complex technique

integral and derivative-based controllers, especially FOPID, has been growing in recent decade. In this context, Muresan et al. [150] investigated this problem and evaluated the adjustments of FOPID parameters. After reviewing various direct and indirect methods for the adjustment of controller parameters and providing examples to investigate the performance of each method, self-adjustment strategies of parameters based on the identification of the controlled system were addressed. Besides, the indirect adjustment of controller and finally their application on the controller were emphasized, as illustrated Fig. 24. However, as an example of non-fuzzy FO schemes, for a spherical robots subjected to saturation constraints as an input, Zhou et al. [151] developed a smooth fractional-order sliding mode control (FOSMC) approach to achieve maximum stability in tracing parallel with mitigating chattering

phenomena, while the fractional Lyapunov approach was utilized for stability rigorously.

Furthermore, Ahmad et al. [152] applied free-model method to design an output feedback controller based on adaptive FO ultra-twisted SMC to control a robotic manipulator. In this method, the feedback was measured only from angle output and robust exact differentiation (RED) was applied to estimate speed and acceleration of the joint using the following equations:

$$\begin{aligned}
 \dot{z}_0 &= -\beta_1 |z_0 - q|^{\frac{2}{3}} \text{sign}(z_0 - q) + z_1 \\
 \dot{z}_1 &= -\beta_2 |z_1 - \dot{z}_0|^{\frac{1}{2}} \text{sign}(z_1 - \dot{z}_0) + z_2 \\
 \dot{z}_2 &= -\beta_3 \text{sign}(z_2 - \dot{z}_1)
 \end{aligned} \quad (3)$$

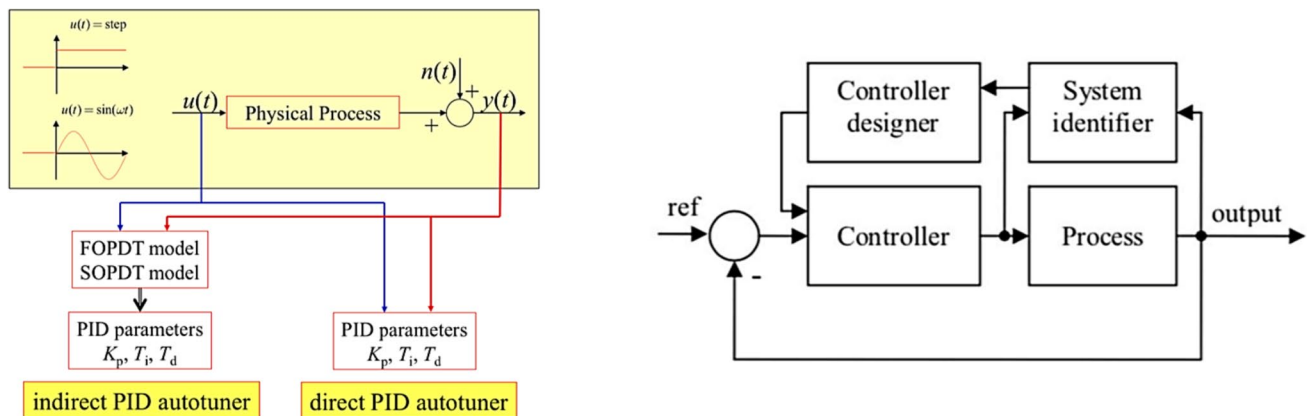


Fig. 24 Direct and indirect methods for adjusting the parameters of PID-type controllers and self-adjustment of parameters [150]

Moreover, Gupta et al. [153] developed a FO PID controller for a robotic manipulator with flexible joint. Besides, feedback control mode was applied with using various Calman-type filter estimators to estimate unmeasured modes. In addition, Zhou et al. [154] proposed FO ding mode control based on convolutional deep neural network (DNN) to control the route of robotic manipulators. In this DNN, estimation of system uncertainties and SMC maintained the stability. Along with preserving stability based on Liapolov method, they conducted a simulation to show that this control approach could effectively eliminate chattering phenomenon and increase stability. Furthermore, Jie et al. [155] studied the problem of route tracking control of a robotic manipulator with seven DOF using FO terminal SMC accompanied by sliding turbulence observer. This observer was able to address chattering problem and enhance the controller convergence using FO derivatives. Besides, the Granold-Letnico method was employed for numerical calculations. Additionally, Chávez-Vázquez et al. [148] reviewed FO controllers in various types of robots and self-driven devices from robotic manipulators and wheeled robots to quadcopters, drones, and submarines using various approaches including PID, sliding mode, backslide, adaptive, fuzzy, and NN. Moreover, Bingi et al. [30] reviewed the application of FO controllers such as PID, FO, fuzzy FO, FOSMC, and FO backslide in robotic manipulators. In this review, more than 100 studies were discussed, in which 60% were related to two-jointed robotic manipulators and in total, 23% were related to one- or three-jointed robots and the remaining studies were related to manipulators with more than three joints, indicating that the application of FO control approaches to industrial robotic manipulators with more than five joints requires more research.

5.1 Fuzzy controllers with FO derivatives

Controllers that employ FOPID strategies combine the advantages of fractional calculus and fuzzy logic to improve control accuracy, dependability, and disturbance rejection capabilities known as fractional order fuzzy proportional-integral-derivative (FOFPID) controller [156]. For instance, a fuzzy fractional order proportional integral derivative (FOPID-FLC) was developed for a lower limb exoskeleton, which showed substantial enhancements in error performance indicators and the robustness of control actions, making it suitable for rehabilitation applications [157]. In this framework, Cao et al. [158] studied robotic manipulators interacting with environment and developed a PID fuzzy adaptive impedance controller with FO to modify contact force overshoot in transition state (contact initiation) and tracking of desired force in stability state (during motion). In this controlling approach, linear controller

with fractional derivative was used to enhance controlling performance and to avoid fractional derivative calculations; the fractional derivatives were transformed to correct derivatives using their natural property through percentage control. Also, to obtain environmental uncertainties, a fuzzy system was applied to adaptively adjust impedance coefficient. This approach could address the limits of traditional approaches and obtain better controlling performance in simulations. For the purpose of tracking performance and ensuring boundedness for uncertain FO nonlinear systems subjected to full-state constraints, recently, Liu et al. [159] presented adaptive fuzzy command filter, where they applied the FO command filters combined with asymmetric barrier Lyapunov function that utilized to perform the stability analysis. Besides the high tracking performance which was proven by two visual examples, the system also handles input constraints and approximation errors. Furthermore, Abdulameer et al. [160] developed four different structures by combining fuzzy PID (FPID) controllers and FO operators to control a manipulator with three joints and used an innovative optimization algorithm to minimize integral time square error (ITSE) to adjust efficiency coefficient. In addition, Izadbakhsh et al. [161] formulated a fifth-order nonlinear coupled problem to control a robotic manipulator considering electrical driving system. Therefore, a fuzzy controller with fractional derivatives was developed to solve this problem in which parts of a dynamic electrical motor were estimated as a mass uncertainty by a fuzzy system and feedback into controlling approach and ultimately, feedback control was calculated using a proportion term along with a Capoto-type derivative term. After proving stability using Liaponov method, the achieved outcomes of the controlling approach were experimentally examined. Likewise, the same authors published further work without using fuzzy system to estimate uncertainties [162]. However, Faieghi et al. [163] stabilized a nonlinear dynamism known as Liu system using FOSMC approach and used fuzzy logic to determine coefficients related to the controller to eliminate chattering. Besides, the Liaponov approach was used to define stability. Furthermore, Ullah et al. [164] applied adaptive fuzzy SMC to investigate a class of nonlinear dynamics with uncertainty. The key purpose of their research was to explore the following second-order differential equations, which were the general form of mechanical system movement including robots:

$$\begin{cases} \dot{x}_1 = x_2 \\ \dot{x}_2 = f(X, t) + g(X, t)u + d(X, t) \\ y = x_1 \end{cases} \quad (4)$$

An integral term was integrated to extend sliding surface and made integral and derivative orders fractional in sliding surface equations, and made it FO. Besides, a fuzzy system

was utilized to compensate system uncertainties neglected in the nominal model and developed adaptive laws to adjust this

fuzzy observer. Ultimately, the FOSMC law was developed and obtained as

$$\begin{cases} u = u_{eq} + u_s \\ u_{eq} = g_n(X, t)^{-1} \left(\ddot{x}_d - f_n(X, t) - \hat{L}(X|\theta) - c_1 c_3 e_1 - c_1 D^\alpha e_1 - c_3 D^{1-\alpha} e_2 \right) \\ u_s = -g_n(X, t)^{-1} k \operatorname{sgn}(s) \end{cases} \quad (5)$$

Moreover, Dabiri et al. [165] developed a controller based on ANFIS to control linear dynamic systems along with a FO oscillator. The FO oscillator introduced as a spring spot, which depends on the order of fractional derivative, could imitate the behaviors of a linear spring ($\alpha=0$) to a linear oscillator ($\alpha=1$) in a continuous spectrum. Furthermore, Duc et al. [166] performed similar study to Ullah et al. [164], with the difference

that nonsingular terminal SMC was made FO along with an adaptive fuzzy system to estimate sign function whose non-continuity created chattering to stabilize the same class of non-linear dynamics of Eq. (4). The fuzzy system was defined to be of T-S-type with singleton fuzzification, multiplication inference motor, central-average non-fuzzification, and expressed for a multi-input/multi-output system as in the next formula:

$$y_j = \frac{\sum_{l=1}^L \left(\prod_{i=1}^n \mu_{A_i^l}(x_i) \right) \bar{y}_j^l}{\sum_{l=1}^L \left(\prod_{i=1}^n \mu_{A_i^l}(x_i) \right)} = W_j^T \Psi(x), j = 1, 2, \dots, m, \Psi_j(x) = \frac{\prod_{i=1}^n \mu_{A_i^l}(x_i)}{\sum_{l=1}^L \left(\prod_{i=1}^n \mu_{A_i^l}(x_i) \right)} \quad (6)$$

where W_j is fuzzy system parameters, $\Psi_j(x)$ is basic functions, and $\mu_{A_i^l}(x_i)$ is membership function of input x_i (fuzzy-fier). They expressed sliding surface and FOSMC based on dynamic stabilization (4) as

$$s = {}_{t_0}D^\alpha e_1 + \frac{1}{\beta} \operatorname{sig}(\dot{e})^\varphi \quad (7)$$

And the final overall control equation became

$$u = \frac{1}{g_n(x, t)} \left(\ddot{x}_d(t) - f_n(x, t) - \frac{1}{\beta} D^{\alpha+1} e(t) |\dot{e}|^{1-\varphi} - (K_1 s + \hat{u}_{FS} + \delta) \right) \quad (8)$$

Here, to eliminate chattering effect, fuzzy system was introduced to estimate sign function; therefore, we have

$$\hat{u}_{FS} = \hat{u}_{FS}(s|W^*) = W^* \Psi(s) \equiv K_2 \operatorname{sign}(s) \quad (9)$$

where W^* is optimal parameters of fuzzy system and is calculated as

$$W^* = \arg \min_{W \in \Omega} \left[\sup_{s \in R} |\hat{u}_{FS}(s|W) - u_{SW}| \right] \quad (10)$$

Additionally, Mirzajani et al. [167] made a chaotic dynamism called Genesio-Tesi system FO and then introduced a T-S-type fuzzy adaptive controller to estimate those nonlinear equations in the equation of controller. Each equation of the assumed differential system was rewritten as

$$D^\alpha x(t) = \sum_{i=1}^r \frac{w_i(z(t)) \cdot A_i x(t)}{\sum_{i=1}^r w_i(z(t))} = \sum_{i=1}^r h_i(z(t)) \cdot A_i x(t) \quad (11)$$

where $x(t)$ is the vector of state variables and A_i are their matrix coefficients in the differential equation system. In fact, the equations are expressed as a linear combination of T-S-type fuzzy law. Control law was also expressed as fuzzy laws as

$$u(t) = - \sum_{i=1}^r h_i(z(t)) \cdot \left(\hat{A}_{ui} x(t) + \hat{B}_{ui} \zeta(t) + K_i x(t) + L_i \zeta(t) \right) \quad (12)$$

where $\zeta(t)$ is the vector including time stimulations on dynamic system and K_i and L_i output coefficients. Also, the following adaptive laws were expressed for $\hat{A}_{ui}$ and $\hat{B}_{ui}$ coefficients as FO:

$$\begin{aligned} D^\alpha \hat{A}_{ui} &= \sigma_A \cdot h_i(z(t)) \cdot x(t) \cdot x^T(t) \\ D^\alpha \hat{B}_{ui} &= \sigma_B \cdot h_i(z(t)) \cdot x(t) \cdot \zeta^T(t) \end{aligned} \quad (13)$$

Moreover, Deng et al. [168] presented an adaptive SMC to track the route of robotic manipulators where efficiency coefficients of sliding surface were adaptively updated using a FO dynamic. Also, a fuzzy inference system was used to estimate the uncertainties of the nominal model of the robotic manipulator. The stability of the developed controller was proved based on Lyapunov stability method. However, Ardeshiri et al. [169] controlled a two-link robotic arm using the combination of PID controller with FO and a fuzzy system. Defining coefficients of the main controller is the role of the fuzzy system which is selected as Mamdani type. The Oustaloup approximation method was implemented in order to compute the transfer function of FO derivatives in Laplace domain. In addition, a

fuzzy-augmented second-order SMC controller was implemented to a twin-rotor physical platform and proposed by Ozer et al. [170]. The main function of this system was to work as an axisymmetric helicopter-like robotic and act under external disturbance. To regulate the sliding surface parameters dynamically, the fuzzy logic was employed, thereby enhancing the disturbance rejection and mitigating chattering properties. Besides, the robustness and improved tracking precision were experimentally validated against conventional SMC technique. Though, the authors developed this system only to exemplify the utilization of hybrid fuzzy-SMC approach in modern robotic control systems. However, Zaway et al. [157] employed the FOPID-FLC to control the motions of a lower body external skeleton robot used in stepping rehabilitation. Fuzzy laws evaluated the ability of the patient in using the rehabilitation robot and then, the respective FO control law was activated based on the output of the fuzzy system. The performance of the control method was evaluated through simulations based on various criteria such as ISE, MSE, and ITSE. Similarly, Wen et al. [171] applied

FOPID-FLC system to control the force of a parallel robot with six manipulators. Figure 25 illustrates the schematic diagram of their developed method. Also, the following equation was proposed for FO to enhance the accuracy of the Oustaloup estimation method:

$$s^\alpha \approx \left(\frac{dw_h}{b} \right)^\alpha \left(\frac{ds^2 + bw_h s}{d(1-\alpha)s^2 + bw_h s + d\alpha} \right) \prod_{k=-N}^N \frac{s + w_k'}{s + w_k}$$

where

$$w_k' = \left(\frac{dw_b}{b} \right)^{\frac{\alpha-2k}{2N+1}}, w_k = \left(\frac{bw_h}{d} \right)^{\frac{\alpha+2k}{2N+1}} \quad (14)$$

Moreover, Sharma et al. [156] analyzed the performance of FOPID-FLC controllers on robotic manipulators. Based on Fig. 26 content, the fuzzy system was applied to estimate controlling function u and its variation rate Δu , and using an FO dynamic, provided a Fuzzy proportional-integral (PI) + fuzzy proportional-derivative (PD) controller which was integrated by the order β and derivatives by the order α . Therefore, the final control signal was

$$\begin{aligned} \frac{T}{(1-z^{-1})} \Delta u(k) + u(k) &= K_e e(k) + K_r \frac{(1-z^{-1})}{T} e(k) \\ \underbrace{\frac{T}{(1-z^{-1})} \beta \Delta u(k)}_{\text{Fuzzy PI}} + \underbrace{\alpha u(k)}_{\text{Fuzzy PD}} &= K_e e(k) + K_r \frac{(1-z^{-1})}{T} e(k) \end{aligned} \quad (15)$$

$$u_{\text{FOFPI}}(k) = [\text{Integrator}]^\beta \beta \Delta u(k) + \alpha u(k) = K_e e(k) + K_r [\text{Differentiator}]^\alpha e(k)$$

This controlling approach was designed for a two-jointed manipulator with each joint controlled separately using this strategy. Additionally, Kumar et al. [172] introduced a new approach known as nonlinear adaptive FOPID-FLC and applied it to control a two-jointed manipulator with dynamic uncertainties, as illustrated in Fig. 27. In this approach,

controlling signal was calculated based on the following FO dynamics:

$$K_1 e(nT) + K_2 (D^\mu(e(nT))) = \underbrace{K_{UPI} (D^{-\lambda}(\Delta u_{PI}(nT)))}_{\text{Velocity form}} + \underbrace{K_{UPD} u_{PD}(nT)}_{\text{Position form}} \quad (16)$$

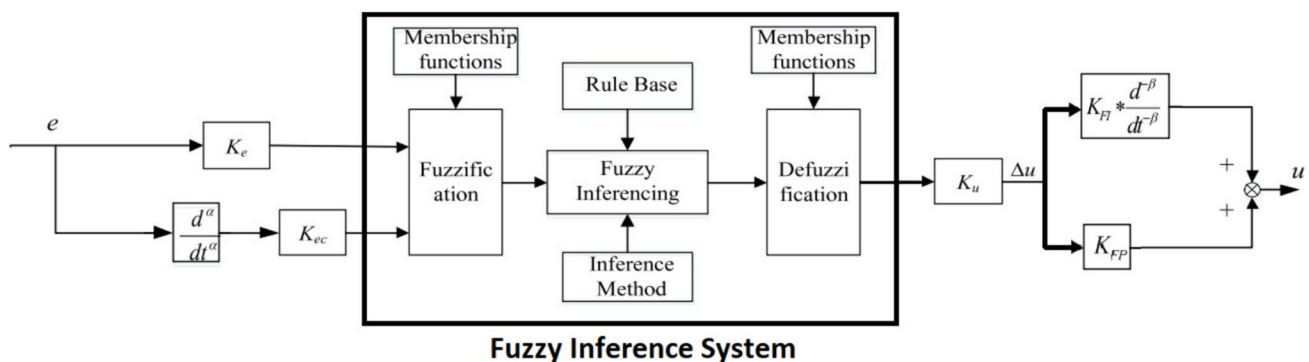


Fig. 25 Fuzzy FO PID control [171]

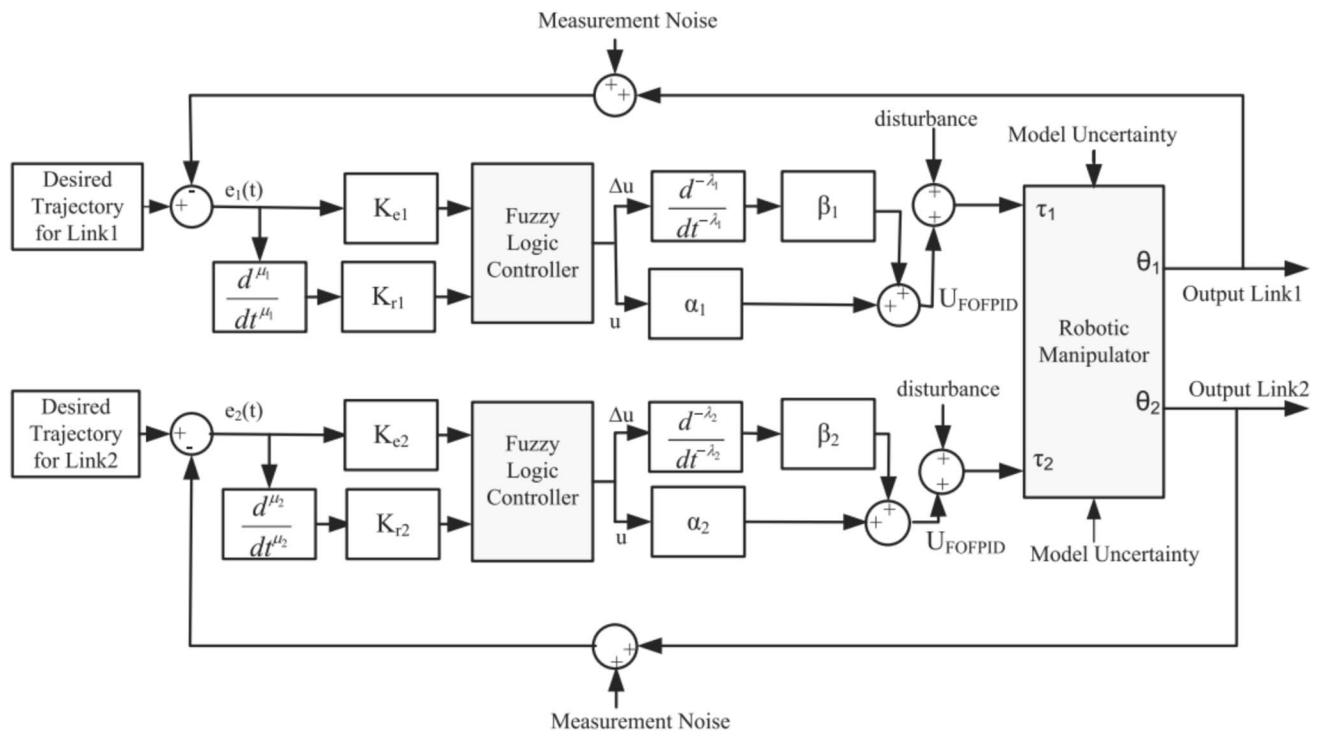
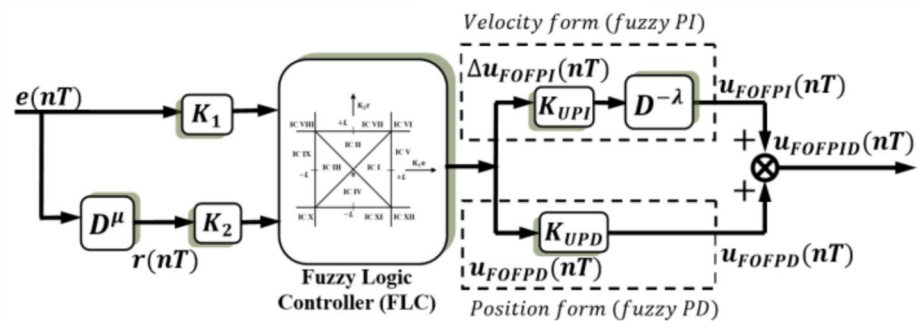


Fig. 26 FOFPID controlling approach applied to a two-jointed manipulator [156]

Fig. 27 Nonlinear adaptive FOFPID controlling approach applied to a two-jointed manipulator [172]



where μ and λ were defined as 1 and the controlling signals u_{PI} and u_{PD} were fuzzy PI and fuzzy PD controls, respectively, and calculated as

$$\begin{aligned} \Delta u_{FPI}(nT) &= K_1 e(nT) + K_2 r(nT) \\ u_{FPD}(nT) &= K_1 e(nT) + K_2 r(nT) \end{aligned} \quad (17)$$

where $r = D(e(nT))$ is variation rate of tracking error. The efficiency coefficients K_1 and K_2 in their controlling approach were evaluated using an evolutionary optimization algorithm, which targets function including total integral time absolute (ITA) of tracking error controlling signal variations of two-jointed manipulator which were weighted. Besides, the Granvold-Latniko law was applied for numerical calculations of FO as

$$D^\beta(f[n]) = T^{-\beta} \sum_{i=0}^{\infty} d_i f[n-i] \approx T^{-\beta} \sum_{i=0}^{100} d_i f[n-i]$$

where d_i is define as

$$d_i = (-1)^i \binom{\beta}{i} = \left(1 - \frac{1+\beta}{i}\right) d_{i-1}, i = 1, 2, \dots, d_0 = 1 \quad (18)$$

Furthermore, Kumar et al. [173] introduced FO fuzzy PI Δ D μ control to track motions in a robot with five DOF. Besides, further changes in the fuzzy system such as changing output membership functions from singleton to triangular and their number from 2 to 7 were accomplished. Additionally, Kumar et al. [174] presented their FOFPID controlling approach to track the route of a three-jointed

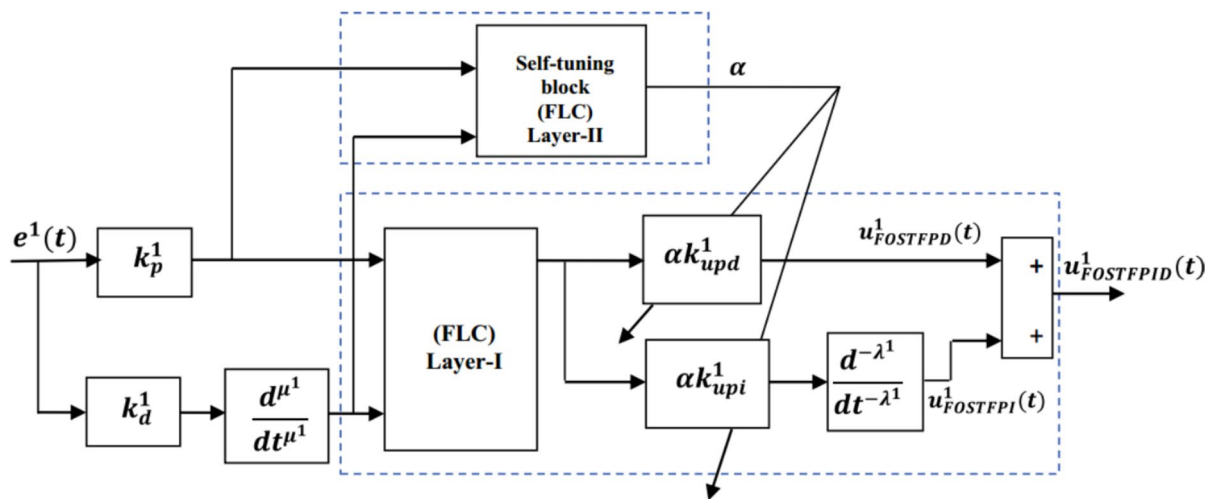


Fig. 28 The diagram of self-adjusting FOFPID controlling approach applied to a three-jointed manipulator [174]

robotic manipulator. In this approach, as seen in Fig. 28, efficiency coefficients of FO PD and PI were optimized by Cuckoo search algorithm and ultimately, calculated the controlling signal as

$$u_{FOSTFPID}^1(t) = \underbrace{ak_{upd}^1 u(t)}_{FOSTFPD} + \underbrace{ak_{upi}^1 \left(\frac{d^{-\lambda^1}}{dt^{-\lambda^1}} \left(\frac{d^{\mu^1} u^1(t)}{dt^{\mu^1}} \right) \right)}_{FOSTFPI} \quad (19)$$

Moreover, Sharma et al. [175, 176] proposed new approaches for FOPID-FLC control to track the motions of a two-jointed robot which included an FO dynamic control and a Mamdani fuzzy system. Besides, they addressed that to determine efficiency coefficients, several novel optimization algorithms such as Cuckoo algorithm [175]

or non-dominated sorting genetic algorithm [176] can be used. Besides, Sharma et al. [177] added a novelty to their previous work by switching between FOPID controller and FOFLC, exploiting the advantages of old and the developed models. As a finding, PID controller could eliminate permanent state error and fuzzy controller had shorter stand-up time and overshoot consideration. Therefore, when error exceeded a threshold level, control was delivered to fuzzy system to minimize it faster, with lower overshooting. Meanwhile, when error was below this threshold level, FOPID controller was activated to maintain permanent error. Determination of the optimum value of this threshold was also performed with Cuckoo search algorithm. Furthermore, Chhabra et al. [178] developed a fractional-order fuzzy PD + I (FOFPD + I) controller and integer-order

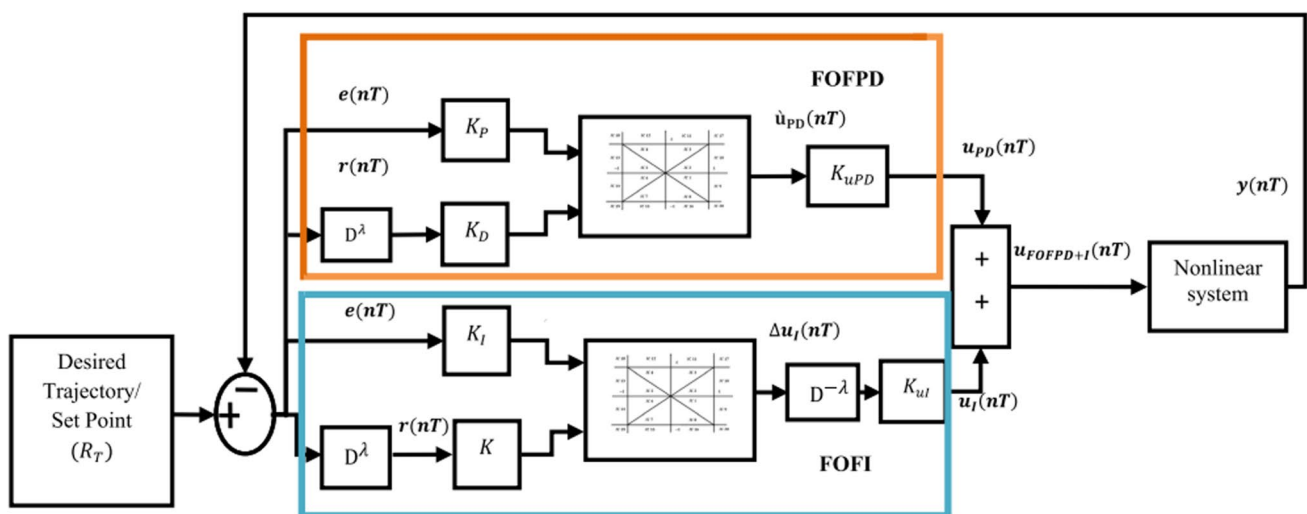


Fig. 29 The diagram of self-adjusting FOFPID controlling approach applied to a two-jointed manipulator [179]

fuzzy PD+I controller (FPD+I) to track the route of a two-jointed manipulator. The controlling approach was simple, as shown in Fig. 29, where controller output was calculated by adding two separate FO fuzzy controllers as

$$\begin{aligned} u_{\text{FOFPD+I}}(nT) &= u_{\text{PD}}(nT) + u_1(nT) \\ u_{\text{FOFPD+I}}(nT) &= K_{u\text{PD}} u_{\text{PD}}(nT) + K_{u1} D^{-\lambda} \Delta u_1(nT) \end{aligned} \quad (20)$$

Muñoz-Vázquez et al. [179] developed an FO PID controller in combination with fuzzy system to track free-model robotic manipulators. In this control approach, a tent space was defined to transform tracking error called s , which was expressed as

$$\begin{aligned} s_0(t) &= K_p \Delta q(t) + K_i \int_0^t \Delta q(\rho) d\rho + K_d \Delta \dot{q}(t) \\ s(t) &= s_0(t) - s_0(0) \Delta q = q - q_d \end{aligned} \quad (21)$$

Then, by defining a T-S-type fuzzy system which received error s and its fractional integral $I^\alpha s$ and created control signal in the presence of uncertainties, they tried to softly tend tracking error on error tent toward origin. The controlling approach is shown in Fig. 30.

Gaidhane et al. [180] used FO fuzzy control to track a two-jointed manipulator in which derivative orders and efficiency coefficients were calculated using an innovative optimization method combining ant colony and grey wolf methods. The ultimate control law (including a few efficiency coefficients) was stated as

$$\begin{aligned} u(t) * &= [G_{PD} G_E] e(t) + [G_{PD} G_{CE}] \frac{d^\mu e(t)}{dt^\mu} \\ &+ [G_{PI} G_E] \frac{d^{-\lambda} e(t)}{dt^{-\lambda}} + [G_{PI} G_{CE}] \frac{d^{\mu-\lambda} e(t)}{dt^{\mu-\lambda}} \end{aligned} \quad (22)$$

Furthermore, Li et al. [181] combined an adaptive SMC based on radius base neural network (RBF) and FO fuzzy system to better estimate nonlinear factors and uncertainties of the system. Besides, a system dynamic based on an FO T-S fuzzy system was modeled and decomposed the uncertainties and disorder of the system to RBF network and used this estimation in SMC law, as stated below:

$$\begin{aligned} u_{eq}(t) &= K_h x(t) - f(t) \\ u(t) &= K_h x(t) - \hat{W}^T \Phi(x(t)) - (\hat{\mu} + \xi) \text{sgn}(s(t)) \end{aligned} \quad (23)$$

Furthermore, Wu et al. [182] designed an adaptive FO non-singular terminal SMC based on fuzzy wavelet neural network (FWNN) to control a wheeled robot moving in all directions. Distinct to its difficult name, it had two distinct parts. The dynamic uncertainties of the robot were estimated using a NN which was, in fact, a linear combination of the wavelet base functions (with transportation and comparison parameters) existing in the fuzzy structure, $f(W, x, c, \varpi) = WT\psi(x, c, \varpi)$, and applied SMC to eliminate external disturbances and follow control order in which FO derivatives were used to accelerate convergence. Moreover, Zhang et al. [183] applied a FOPID-FLC controller to

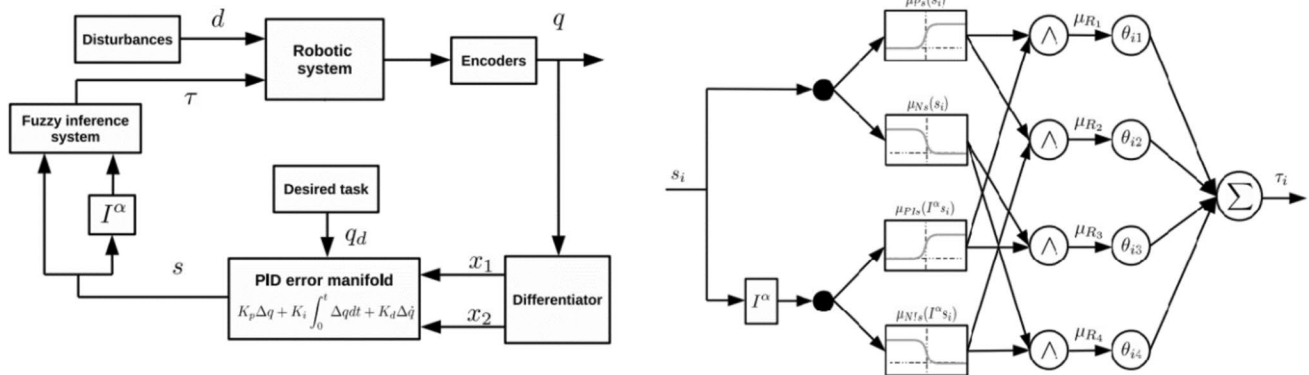


Fig. 30 FO fuzzy system controlling approach [171]

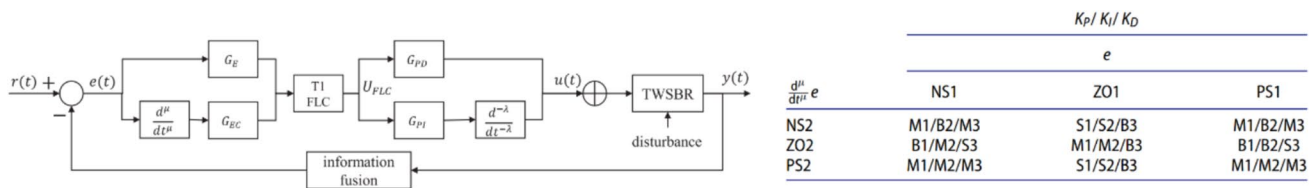


Fig. 31 FOPID-FLC controlling approach and fuzzy law table to determine efficiency coefficients [183]

Damani et al. [185] applied fuzzy FO integral SMC to create an inter-group moving arrangement of wheeled robots. In this controlling approach, application of FO arithmetic-enhanced system robustness against external disturbance and fuzzy system eliminated unfavorable chattering effects. However, a hybrid FO type II fuzzy PID controller was developed by Wang et al. [186] for fixed-wing fuselage attitude control. This novel system considered optimum for yaw and pitch stabilization, and showed significant tracking performance and disturbance rejection. Meanwhile, Xian et al. [89] used the bat optimization algorithm (BOA) to adjust the parameters of these controllers, improving their effectiveness in real-time applications. Besides, Parikh et al. [187] employed a blend of advanced controllers, such as fuzzy logic and FOPID controllers, in six DOF robotic arm to support individuals with disabilities in feeding themselves independently. This configuration reduced positional inaccuracies and overshoot, guaranteeing precise and effective food delivery.

Fractional derivatives enhance the HRI modeling within type II fuzzy systems by introducing adaptive features through their nonlinear gains, maintaining the linear framework of controllers, and providing improved performance compared to conventional PID controllers [89]. The FOFPD + I controller, which is based on fractional-order fuzzy proportional derivative (FOFPD) and fractional-order fuzzy integral (FOFI) controllers, achieves over 80% efficiency in comparison to traditional PID and fuzzy logic controller (FLC) [188]. Although, the proposed controllers display remarkable simulation results, real-world implementation and practical-use issues require further investigation [188]. Furthermore, Kumar et al. [189] applied a fractional derivative PD + I controller in combination with type II fuzzy system to stabilize chariot and pendulum system, as presented in Fig. 33. Besides, the ABC optimizer was employed to determine input and export efficiency coefficient values of the fuzzy system.

Moreover, Zakeri et al. [43] developed an FO super-twisted SMC along with an optimum interval type II fuzzy system to control half-driven and full-driven nonlinear systems. The controller was tested in the presence of uncertainties such as inverse pendulum and ball and rod and claimed that their method significantly decreased tracking error as well as controlling effort and especially chattering. Besides, it was suggested that the significant

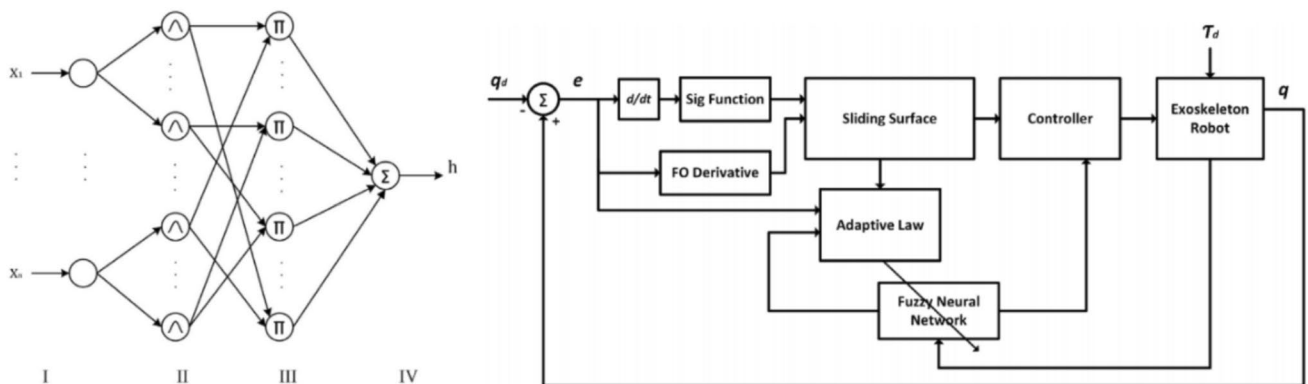
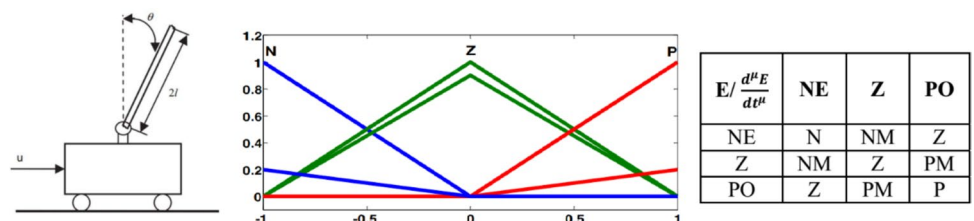


Fig. 33 Membership functions of type II fuzzy system and their law tables based for the stabilization of chariot and pendulum system [189]



results can be due to the combination of several methods in a single algorithm. The stability of the algorithm in limited time was proven using Liapanov method. In more detail, the sliding surface was assumed as a combination of error FO derivative and the type II fuzzy system as

$$s = {}_CD^\alpha \dot{q} + \lambda_0 F_{IT2F}(\eta), \quad \eta = \lambda_1 q \quad (24)$$

where λ_0 and λ_1 are positive coefficients which are multiplied with the input and output of the assumed fuzzy system. Three controlling approaches were proposed in this research. The first approach was a super-twisted SMC. The second one was an FO super-twisted SMC. The comparison of the two approaches is illustrated in Table 2. While the third approach was an interval type II FO super-twisted SMC.

Furthermore, Jamshidi et al. [44] proposed an interval type II FO PID controller to track the route of a two-jointed robot and evaluated its performance through simulation, as illustrated in Fig. 34. In this approach, controller coefficient was determined with whale optimization algorithm.

Moreover, Shi [45] proposed interval type II FO PID control to control a chariot and pendulum system in the existence of parametric and structural uncertainties. Besides, the system performance has been evaluated by simulation. Else, it was claimed that this approach enhances the controlling performance in terms of decreasing overshoot and accelerating response and stabilization time. Additionally, Azar et al. [190] combined PID, SMC, and interval type II FO fuzzy compensator controllers to control the motions of robotic manipulator which were considered to compensate model uncertainties. The controlling law was presented as follows and proved its stability based on Lyapunov method:

$$\begin{aligned} \tau_q &= D(q)(\tau_{eq} + \tau_{sw} + \tau_{f2}) \\ \tau_{eq} &= \left(k_{D2} \frac{d^{\mu_2}}{dt^{\mu_2}} + k_{I2} \frac{d^{-\lambda_2}}{dt^{-\lambda_2}} + k_2 \right) e_2 \\ \tau_{sw} &= k_{sw} \text{sign}(S_2) \\ \text{where : } S_2 &= e_2 + k_{D2} \frac{d^{\mu_2-1}}{dt^{\mu_2-1}} e_2 + k_{I2} \frac{d^{-\lambda_2-1}}{dt^{-\lambda_2-1}} e_2 + k_2 \int_0^t e_2(\tau) d\tau \\ \text{Fuzzy type II : } \tau_{f2} &= \xi^T \Theta \end{aligned} \quad (25)$$

Table 2 Comparison of super-twisted SMC and FO super-twisted SMC [43]

FO super-twisted SMC	Super-twisted SMC
${}_CD^\beta(z) = \varphi(z, t) + \mu, 0 < \beta \leq 1$ $ {}^{RL}_0 D^{2-\beta} \varphi(z, t) \leq \delta$ $\mu = I^{1-\beta}(v_n + v_s)$ $v_n = -k_n z ^{\frac{1}{2}} \text{sign}(z) - k_{I1} z$ $\dot{v}_s = -k_s \text{sign}(z) - k_{I2} z$	$\dot{z} = \varphi(z, t) + \mu$ $ \dot{\varphi}(z, t) \leq \delta$ $\mu = v_n + v_s$ $v_n = -k_n z ^{\frac{1}{2}} \text{sign}(z)$ $\dot{v}_s = -k_s \text{sign}(z)$

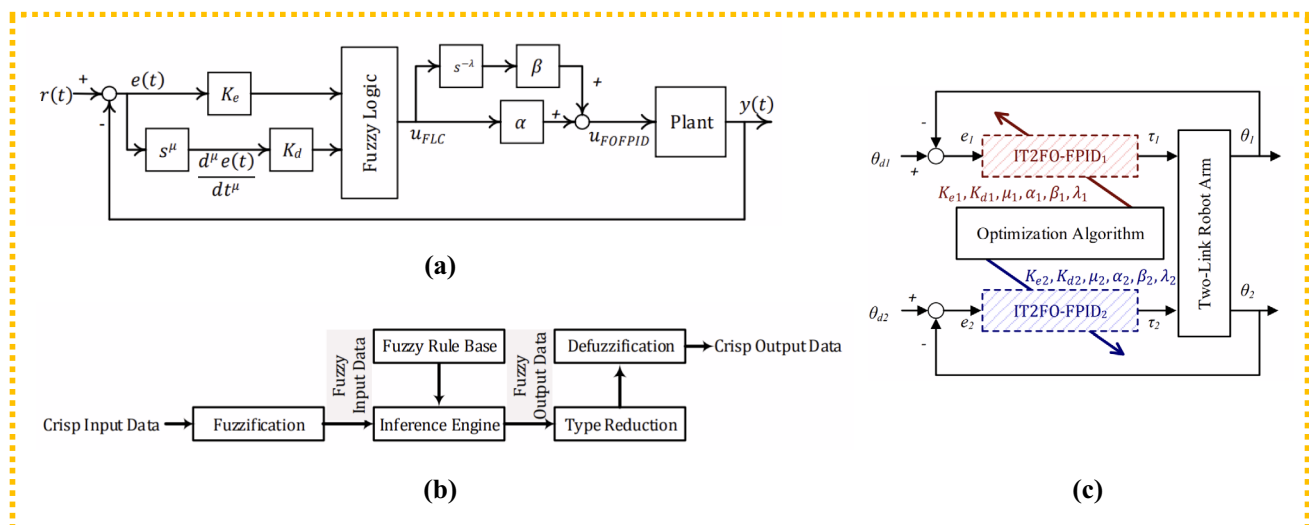


Fig. 34 Schematic of block diagrams for the optimized interval type II FO PID control to track the route of two-jointed robot system: **a** closed-loop system with FO-FPID controller, **b** T2FS configuration, and **c** with IT2FO-FPID controller [44]

Therefore, the SMC efficiency and fuzzy system parameters should be adaptively updated according to the following dynamics:

$$\begin{aligned} \dot{k}_{sw} &= \frac{1}{k_{sw}} S_2^T \left(\left(\text{sign}(S_2) + \xi^T \Theta - \dot{x}_{2d} \right) - D^{-1}(q)(C(q, \dot{q})\dot{q} + g(q)) \right) \\ \dot{\Theta} &= -\frac{\xi}{\|\xi\|^2} \xi^T \Theta \end{aligned} \quad (26)$$

Patel et al. [46] proposed mathematic expansions for a robustness controller for a nonlinear system with uncertainties based on FO interval type II fuzzy system, in which the discharge of cascade double tanks and the suggested method were defined by simulations. The evaluation and the proof of controller stability were accomplished by Laplace space. Besides, to optimally adjust parameters (including efficiency coefficients and orders of fractional integral and derivative), innovative methods such as flower pollination method was applied. The general approach of controller and their association functions type II fuzzy system inputs (error and error variations) are shown in Fig. 35.

5.3 Variable impedance/admittance control with FO derivatives

VIC offers potential uses in enhancing compliance during human–robot interactions, especially in situations involving external force disturbances, which aids in the safety and

adaptability of cross-user human–robot systems [191]. To the best of the knowledge of the authors, in 2017, Aydin et al. [192] first introduced the idea of VAC with FO dynamics for physical HRI. Their controlling approach much resembled traditional methods for admittance control with the only difference that $Y(s)$ admittance model, which determined reference velocity v_{ref} in terms of desired force F_{ref} , was an FO model, as illustrated in Fig. 36. They claimed that the FO admittance model was much more effective than the normal model against impedance variations of human arm. In their following research in the next year [193], they described that if the impedance model was assumed as $Z_{\text{FOAC}}(s)$, where $0 < \alpha < 1$ is the order of fractional derivative, the equation could be rewritten in frequency domain as

$$Z_{\text{FOAC}}(j\omega) = m_F(j\omega)^\alpha + b_F = b_F + m_F \omega^\alpha \cos\left(\frac{\alpha\pi}{2}\right) + jm_F \omega^\alpha \sin\left(\frac{\alpha\pi}{2}\right) \quad (27)$$

It was seen from the equation that effective mass $b_F + m_F \omega^\alpha \cos\left(\frac{\alpha\pi}{2}\right)$ and effective oscillation $m_F \omega^{\alpha-1} \sin\left(\frac{\alpha\pi}{2}\right)$ in the fractional model were more flexible than the normal model, in which the corresponding values were b_F and m_F , since it depends on working frequency ω and coefficient α .

For the purpose of enhancing force tracking on uncertain constant surface, Wang et al. [194] proposed an admittance controller with simulated delayed resonator. By integrating the delayed resonator, it caused to overpower the controller

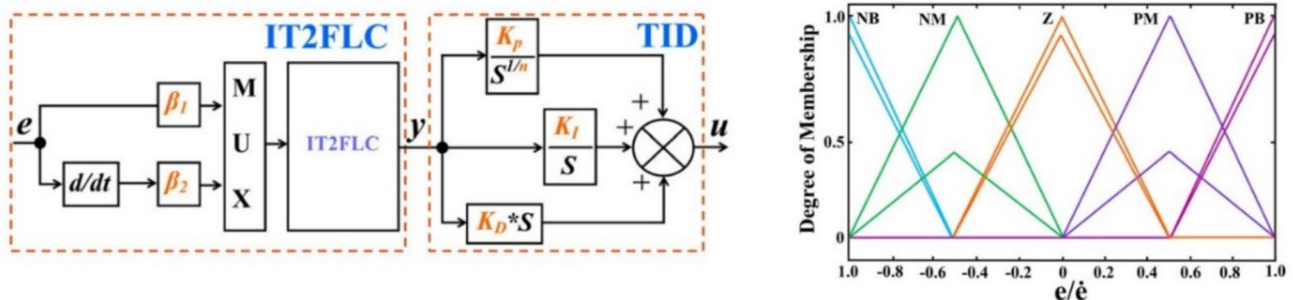


Fig. 35 The diagram of interval type II FO PID control and membership functions of fuzzy system [46]

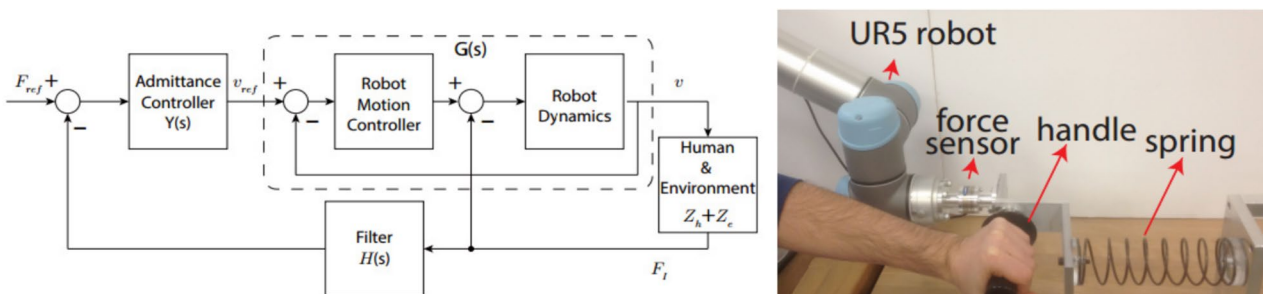


Fig. 36 Diagram of VAC for physical HRI [192]

oscillations and enhance force stability. Moreover, Sirintuna et al. [195] extended their previous approach to a variable FO admittance control to obtain more adaption with the task. The aim of this research field is guiding robotic manipulator in a drilling task, and to do so, they have considered derivative order coefficient α in admittance model in respect to the distance of the tool from variable working location; i.e., initially, when the tool was further than a certain distance to working location, $\alpha = 1$ was considered and the time of contact with the working part and drilling was linearly decreased with distance. By proving stability, they showed that the decrease of coefficient α increased the stability and robustness. Meanwhile, a fractional order admittance controller (FOAC) with five parameters was developed by Wu et al. [196], which included a rear fastest tracking differentiator to overpower interaction impacts through robotic precision polishing. Besides, the Pareto multi-objective optimizer was utilized for tuning the controller parameters by considering frequency- and time-domain performance indices. The experimental results revealed to be in good agreement with simulation, and a remarkable enhancement in force control precision and the error drop from 15 to 5%. However, Chen et al. [31]

investigated the idea of FO VIC in robots and through several simulations, investigated the effects of integral and derivative order on absorption (increase), preservation, or elimination (decrease) of system energy in making smooth and easy physical contact. They did not consider impedance variation in their research and investigated the responses of the system by simulations only considering the variations of coefficients and fractional derivative in impedance model. Furthermore, Ma et al. [197] presented a controlling technique for telerobotic systems based on admittance control in which dynamic model uncertainties were adapted with a NN and route tracking was performed by FOSMC. Although this impedance model was based on a linear model with correct derivatives, SMC and adaptive law of NN Bruzzone et al. [198] developed a FO impedance control approach named KDHD for Steward platform parallel robots. In this method, the relation between extended force as well as the position and position error of end effector formed the impedance model. The main idea of this research was to consider a half-order (HD) derivative along with the terms of oscillation (D) and hardness (K) for six DOF non-redundant parallel robot, which stated the final form of impedance model as

$$\tau = \left(\mathbf{J}_p^T(\mathbf{q}) \right)^{-1} \left(\left(\mathbf{K}_{KDHD} - \frac{W_{1/2}(n)}{T_s^{1/2}} \mathbf{HD}_{KDHD} \right) \begin{bmatrix} \mathbf{x}_d - \mathbf{x}(\mathbf{q}) \\ \mathbf{E}_d - \mathbf{E}(\mathbf{q}) \end{bmatrix} + \mathbf{D}_{KDHD} \begin{bmatrix} (\mathbf{x}_d - \mathbf{x}(\mathbf{q}))^{(1)} \\ (\mathbf{E}_d - \mathbf{E}(\mathbf{q}))^{(1)} \end{bmatrix} + \mathbf{HD}_{KDHD} \begin{bmatrix} (\mathbf{x}_d - \mathbf{x}(\mathbf{q}))^{(1/2)} \\ (\mathbf{E}_d - \mathbf{E}(\mathbf{q}))^{(1/2)} \end{bmatrix} \right) + \tau_g(\mathbf{q}) \quad (28)$$

Ding et al. [199] presented a FO impedance control for the interaction of robotic manipulator with the environment where only the derivative of oscillation term was FO, as presented in Fig. 37. They showed that compared to normal impedance model, after reaching stability and based on ITSE criterion, their approach performed more successfully. In addition, Ding et al. [132] extended their previous work (human–environment interaction) by using the same FO impedance model with 0 hardness coefficient and FO oscillation term derivative only by adding a compensator called “dynamic feed forward” to robot controller which resulted in performance improvement.

6 HRI/HRC under variable impedance/admittance control

Bian et al. [200] applied the concept of “dynamic system” in designing a VAC in physical HRI. Dynamic system is a concept for robot route planning, and it was introduced into the admittance model to simultaneously obtain desired route and smooth motion in the controlling algorithm. Admittance variation in this approach was realized by an artificial oscillation field. Besides, Bae et al. [201] developed and tested a VAC for the collaboration of a human and a hydraulic robotic manipulator. The design was simple and include the

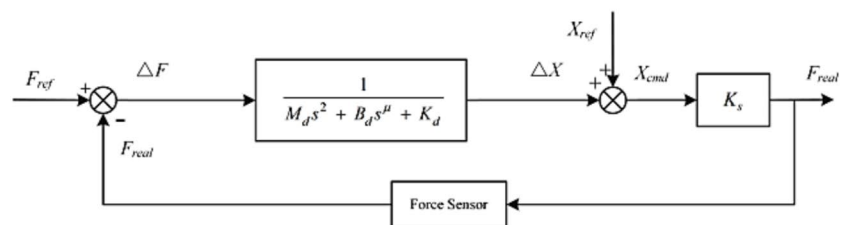
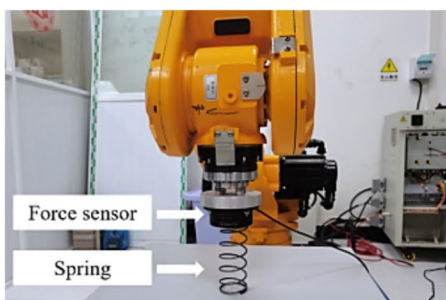


Fig. 37 Diagram of FO VAC for physical HRI [199]

following parts: a classifier estimated whether the human intends to perform a free motion or a precise one, oscillation parameter has been adjusted based on the estimation of the intention of the human motion, and balance point of robot while following human motion was determined using the concept of virtual hardness guidance. Furthermore, Tao et al. [202] developed a safe admittance control approach for rehabilitation exercises of human hands and arms. In this approach, the safe motion domain of exercise was determined based on a space classifier model which used the super points obtained from the biomechanical model of arm to determine safe, hazardous, and interactive subspaces. Then, hardness parameter in admittance model was adjusted based on space classifier model to obtain a safe HRI during exercise, as presented in Fig. 38.

Moreover, Wang et al. [203] proposed a VAC for HRC with the aim of being as similar as possible to human in human–human collaboration. This concept was applied to adjust admittance model parameters based on the indexes of route smoothness and average consumed energy. In addition, Mujica et al. [204] proposed a robust VAC approach for HRC in the transportation of objects with unknown weights, as presented in Fig. 39.

Furthermore, Grafakos et al. [205] applied electromyography signals of arm muscle to smoothly move a robotic manipulator on a desired route. In this approach, oscillation of impedance model was switched based on a threshold level of muscular activity between two values. Meanwhile, Wu et al. [206] applied the concept of learning based on observation and determined the hardness of impedance model by measuring electromyography (EMG) signal of arm muscle to perform a smooth motion in HRI. Moreover, Sirintuna et al. [207] adjusted inertia and oscillation parameters based on motion intention of the operator through designing a classifier which estimated motion intention of the operator based on EMG signals obtained from the arm muscles of the operator and prevented unwanted deviations of the operator arm while working. However, the composite learning-based VIC (CL-VIC) utilizes a multi-mode adaptive control strategy to attain target impedance in parametric uncertainty, guaranteeing seamless transitions and stable interactions [208]. In the context of HRC with variable impedance or admittance control, one benefit is improved adaptability to human intentions and environmental changes [209]. Simultaneously, a significant challenge lies in real-time adaptation to human and ecological forces [210].

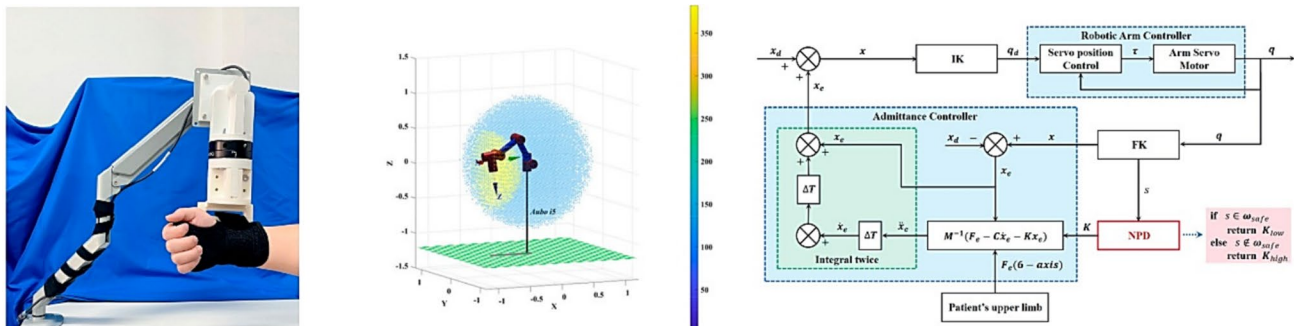


Fig. 38 Diagram of control approach and experimental setup of HRI with the idea of safe motion space classifier [216]

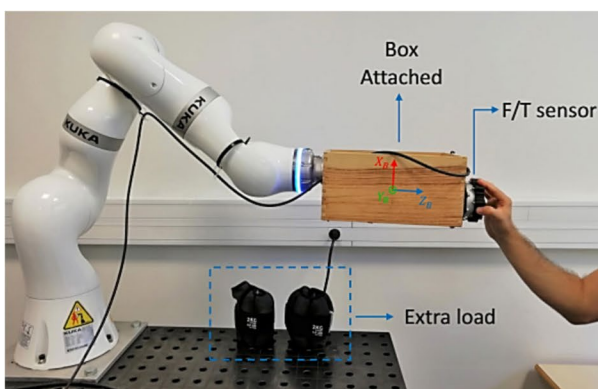


Fig. 39 Diagram of control approach and experimental setup of HRI with the idea of robust VAC [204]

6.1 Control with fuzzy systems

VIC adjusts a robot's stiffness and damping settings dynamically to enhance interaction compliance and safety. Fuzzy systems are often utilized to modify these settings based on interaction forces and position and velocity discrepancies [211]. VAC alters the robot's reaction to external forces by changing real-time parameters such as damping and stiffness. Fuzzy systems improve these modifications, making them more intuitive and responsive to human intentions [212]. These systems can precisely adjust the robot's admittance damping using expert knowledge, facilitating smooth cooperation and boosting safety during joint tasks [212]. Besides, the trials with collaborative robots have demonstrated that fuzzy VAC can effectively and dynamically modify damping coefficients, and enhance compliance and motion performance in physical interactions [212]. Furthermore, fuzzy systems can adjust to the stiffness of the operator's arm, ensuring the robot's actions are in harmony with human intentions [212]. However, studies involving various robots, such as the UR3e and KUKA LWR, indicate that fuzzy VAC techniques significantly improve the compliance and safety of human–robot interactions [212, 213]. To enhance safety and adaptability in HRI system during gait assist, a developed single-input adaptive fuzzy sliding model control was presented by Jin et al. [214] for lower extremity exoskeleton, in which the interaction torque between human and robot was estimated and regulated. Besides, to ensure the stability under time variation step patterns and external

disturbances, the Lyapunov-based adaptation law was implemented. The outcomes revealed that such hybrid system can remarkably minimize joint interaction forces and decrease the tracking error more than traditional fuzzy systems. Furthermore, Dimeas et al. [143] presented a fuzzy VAC approach for HRI in which two fuzzy systems were implemented, one for adjusting impedance model parameters and the other for adjusting the database of the first fuzzy system to obtain final robot motion with minimum jerk. Besides, the learning of the second fuzzy system was performed based on speed difference between reference and interaction points. Also, the reference route was calculated based on motion with minimum jerk. Nonetheless, Sun et al. [215] presented an adaptive fuzzy control technique for FO non-self-driven system where Q-learning was applied to adjust the weights of the controller. Besides, the T-S-type fuzzy system was applied to estimate non-measurable states of the system. In addition, Mao et al. [216] used a fuzzy system for adjusting oscillation parameters of impedance model in HRC, as illustrated in Fig. 40. Also, recurrent minimum square error estimates human arm hardness, in which an estimation for the motion intention of the operator was obtained and robot motion was adjusted based on that estimation. The robotic polishing system illustrated in Fig. 41 was developed and proposed by Li and Lian [209], in which they employed the adaptive impedance control based on visual guidance (AICVG). For geometric perception, the AICVG technique integrates real-time visual feedback, while the impedance parameters were dynamically adjusted by fuzzy

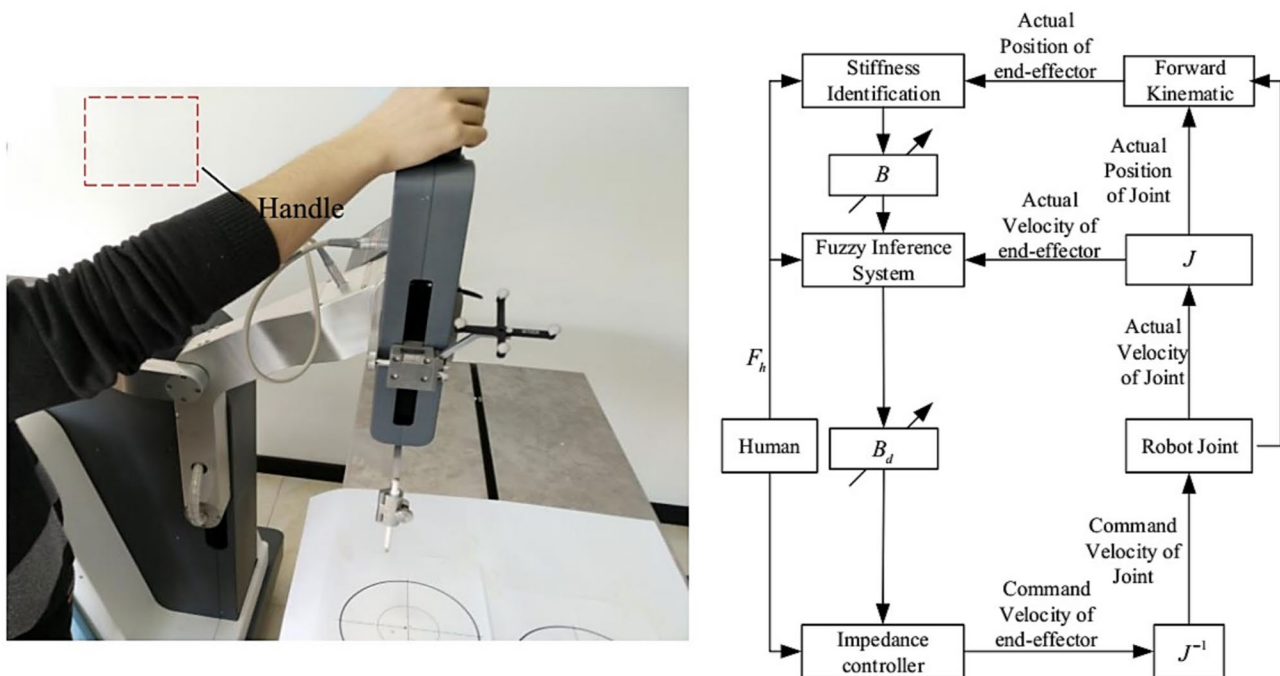


Fig. 40 Diagram of control approach and experimental setup of HRI with the idea of VIC by identifying human arm hardness [208]

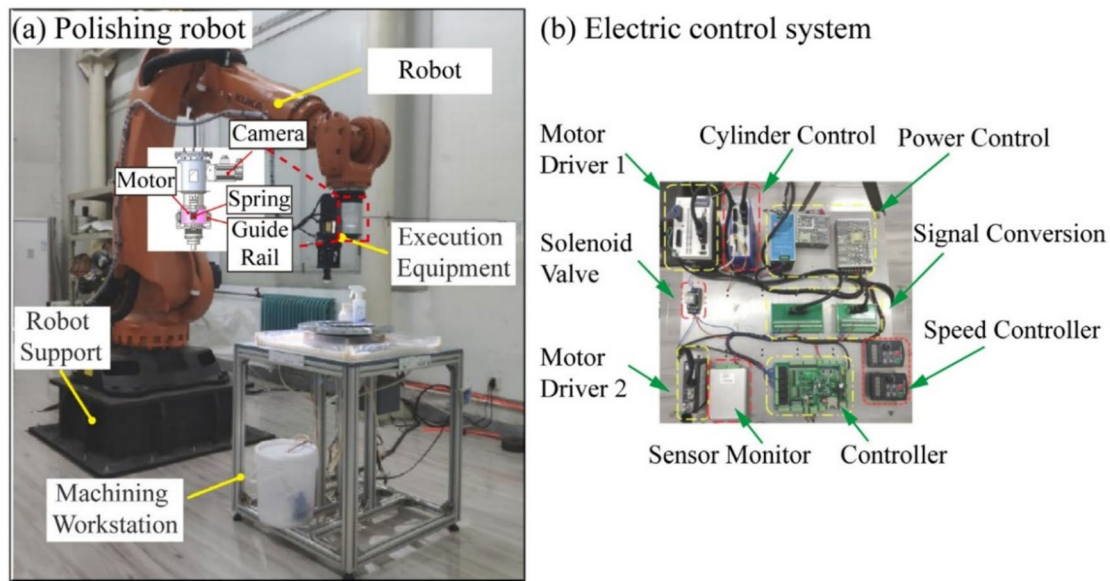


Fig. 41 The experimental platform of KUKA KR16 robotic system for polishing: **a** spatial layout and **(b)** the main components of the electrical control system [217]

logic combined with adaptive tool path generation. It was revealed that the AICVG extremely outperform traditional fuzzy logic-based adaptive and conventional impedance controls. Besides, under dynamic stiffness conditions, the error of maintenance force remains within the range of ± 1.5 N.

Additionally, Toan et al. [218] proposed a fuzzy system to eliminate the necessity of identifying inertia and oscillation parameters in HRI to minimize collision risk of robot and operator. In other words, the replaced admittance model with a fuzzy system calculated the reference motion speed by

receiving data from force-torque sensor and power transmission system. Recently, a robust control approach that integrated SMC, with FOFPID with hybrid swam optimization, was proposed by Karahan and Karci [219]. The approach achieved high-precision control in a six-DOF robotic manipulator under uncertain nonlinear environment. The purpose of FO was to improve robustness and adaptability, while the controller gain was dynamically adjusted by the fuzzy inference. Experimental validations revealed robust performance in disturbance rejection and trajectory tracking, making this

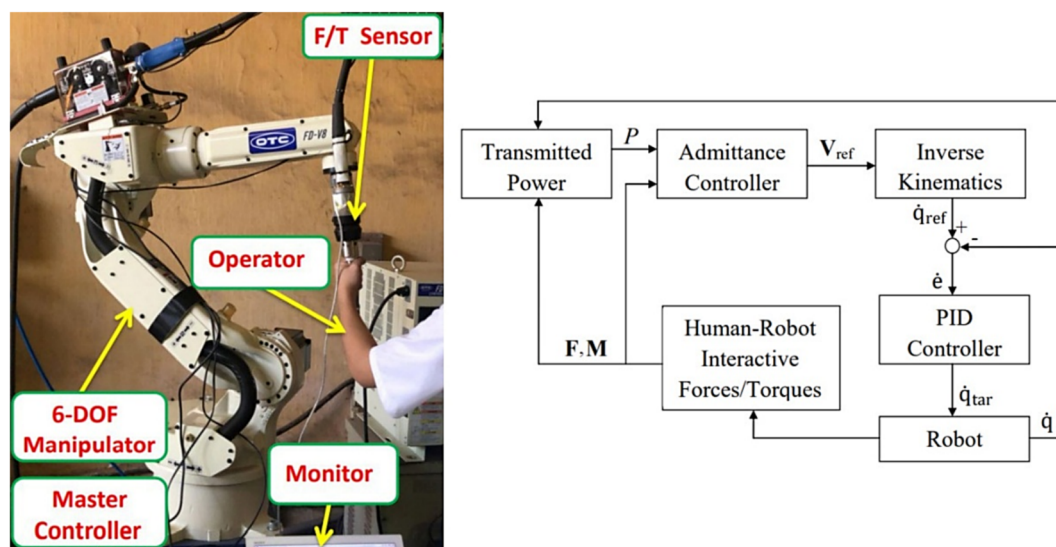


Fig. 42 Diagram of control approach and experimental setup of HRI with the idea of fuzzy-neural admittance with Hedge-Algebras [220]

approach a benchmark for hybrid fuzzy systems. Moreover, Toan et al. [220] developed a VAC by combining a fuzzy system with special calculations called hedge algebras and a perceptron multi-layer NN to create a natural and safe motion during HRI, as presented in Fig. 42.

Furthermore, for collision safety, travel distance, and rotational alignment in a mobile robot for realistic path planning, Kim et al. [221] employed fuzzy analytic hierarchy process (FAHP), in which multi-objective preference was encoded through fuzzy weights and hierarchical decision layers. The results revealed that FAHP was more suitable for mobile robots than typical analytic hierarchy process (AHP). Meanwhile, Li et al. [222] presented a fuzzy adaptive admittance control to track robotic force in an unknown environment, as presented in Fig. 43. Besides, the stiffness parameters were adjusted as in real-time, and the fuzzy logic loop addressed the overshoot at the initial stage of force tracking. This idea could successfully obtain robot interaction with unknown and variable environments.

Additionally, Chien et al. [223] applied a fuzzy system with a network of radius base functions as impedance compensator in physical HRI. This fuzzy-neural system estimated human hand impedance followed with robot motion according to an impedance model which was modified by this estimation. Besides, to estimate the motion intention of the operator, another NN was used which determined reference route in impedance model. Meanwhile, Ying et al. [224] developed a fuzzy admittance system to estimate the motion intention of the operator that resulted in improving efficiency of impedance control in HRI. The multiplication of force to speed of end effector at interaction point was considered as well as their angle as the inputs of their fuzzy system to obtain oscillation parameter value in admittance model as an output. Moreover, Wang et al. [102] used

a fuzzy system to estimate operator motion intention and determination of admittance model in HRI. In addition, a fluctuation observer was used to decrease high-frequency fluctuations due to instability caused by operator arm hardness, as presented in Fig. 44. Moreover, Liu et al. [23] applied a fuzzy system in combination with an adaptive VIC for HRI, in which the impedance control adaptively managed unknown environmental conditions using a NN-and fuzzy system–determined impedance model parameters. Additionally, to improve the trajectory tracking of uncertain lower-limb exoskeleton (LLEX) robot, Alsaied and Li [225] developed new nonlinear technique based on time-delay estimation (TDE) integrated with optimized II fuzzy logic control (T2FLC). The T2FLC structure was optimized by using particle swarm optimization algorithm (PSO). The developed T2FLC-TDE technique accomplished supreme tracking accuracy with fast convergence time compared with conventional PID and time delay based on I fuzzy control technique. Besides, the absolute error of integral time was 0.0011 for knee joint and 0.0021 for hip joint.

6.2 Control with learning methods

Recent progress in learning-based control has empowered real-time adaptability in HRC. For instance, Rahmati et al. [226] implemented reinforcement learning and predictive modeling to accomplish dynamic role adaptation and sustainability optimization in smart manufacturing scenarios. Besides, a Bayesian model has been utilized for path planning for energy-efficient and waste reduction. The results revealed high performance and productivity. While, Gribovskaya et al. [227] investigated collaboration of two robotic manipulators with learning method and variable impedance. It was claimed that this approach could be

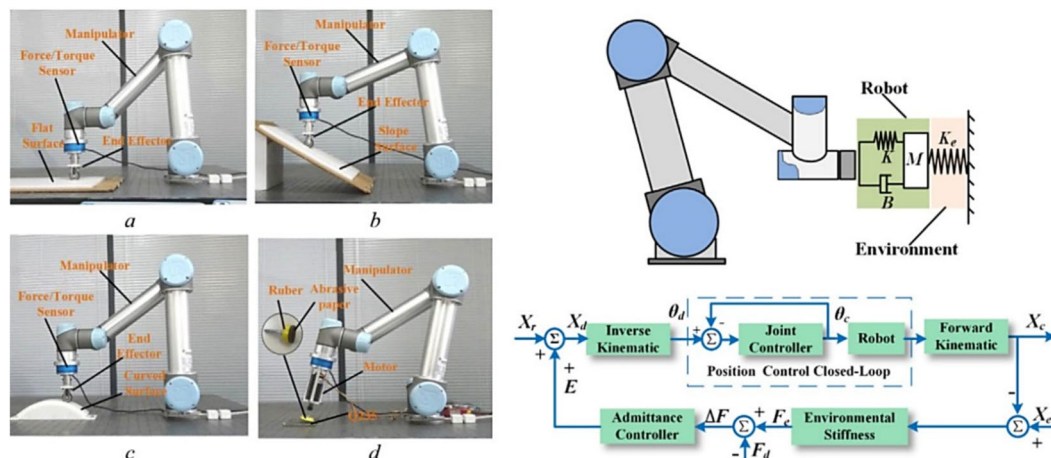


Fig. 43 Diagram of control approach and experimental setup for force tracing in the contact of robot and environment with the idea of fuzzy adaptive admittance control [222]

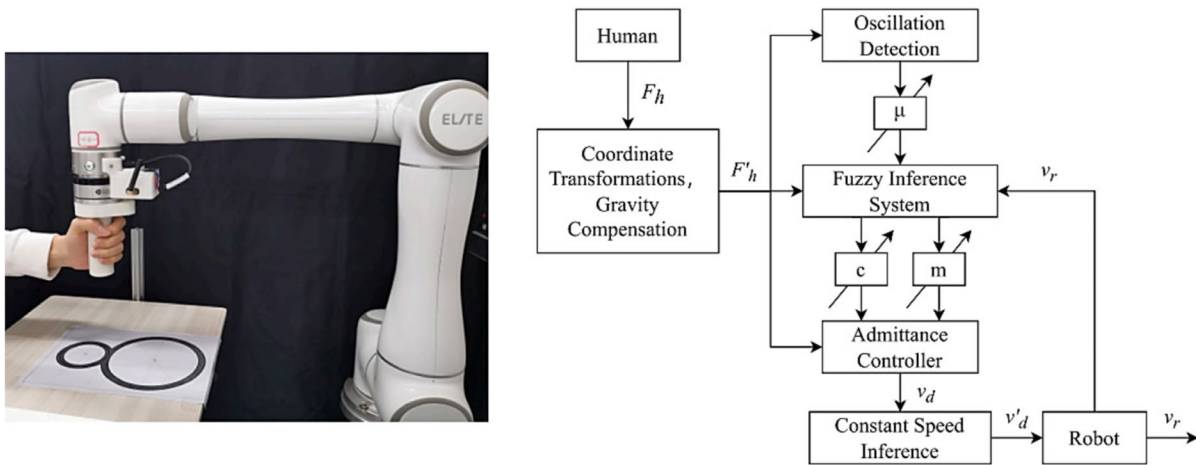


Fig. 44 Diagram of control approach and experimental setup of HRI with the idea of fuzzy VAC for interactive stabilization [102]

extended to HRI, and it was mentioned that by replacing one robotic manipulator with human, the findings of this research could be applied in HRC. Besides, in this approach, one of the robotic manipulators was considered the leader and the second robotic manipulator was considered the follower, and both should collaborate in moving a rod. In fact, the follower robot had to estimate the motions of the leader robot and follows its path. Also, the follower robot had to adjust its inertia and hardness (impedance) through an adaptive learning based on observing the motions of the leader robot to perform a smooth motion. However, each of these tasks was performed via separate learning algorithms. To resolve complications that arise in automated fiber placement (AFP) illustrated in Fig. 45 (a), Villa-Tiburcio et al. [228] developed a novel intelligent control algorithm based on the concept of back-propagation neural networks (BP-NNs) that used to evaluate and compensated online the dynamic differences of AFP process. Besides, for the force/position control in the manipulator robots, the PID/PI control systems were combined to BP-NNs. The obtained results by MATLAB-Simulink revealed that the system achieved high performance. The root mean square error (RMSE) was used as an error metric and seen that the error drops to less than 5% for complex trajectory forces and tacking. Figure 45 demonstrates the entire process from designing the schematic of APF to analyzing and validating method.

Li et al. [229] investigated the interaction of a humanoid robotic manipulator with human by determining impedance model coefficients via learning-based methods. In this approach, a desired interaction was realized through minimizing a target function. To assist physical therapists and support rehabilitation, Amiri et al. [230] developed a swarm-initialized adaptive (SIA)-based controller that consist of adaptive Lyapunov-based controller that updated gains in real time based on user dynamic, while the swarm beetle

antenna searching (SBAS) was used for tuning parameters which enhanced system responsiveness. High accuracy and performance in robustness and steady-state error are suitable for a lower limb exoskeleton for enhancing human motion. Furthermore, Roza et al. [231] considered training robots to collaborate with human by observing human's operation to investigate HRC in replacing heavy parts in industrial factories, in which the probabilistic learning method and dynamic systems were combined to code robot motions during the task. Besides, the force required by the robot to move the object was defined. The main concept depends on virtually assuming that the end effector of the robot was connected to a series of springs such that they performed robotic functions. Additionally, in healthcare sector, specifically for a surgical robot (with two arms), Xie et al. [232] demonstrated a hybrid control strategy that consists of multilayer perceptron (MLP) NN to determine the forward kinematics in real time, and for the trajectory generation, the non-uniform rational B-spline (NURBS) was implemented to improve surgical precision and motion flexibility. In consequence, a smooth curvature-conforming track was granted that is suitable for sensitive manipulation. Such nonlinear hybrid control system based on neural-enhance can be considered a suitable alternative rather than conventional FO and fuzzy impedance models in high-complexity sensitive HRI systems. Also, it eliminates the necessity of physical force sensors. Moreover, Modares et al. [142] developed an assistive control approach based on supportive learning to reduce the amount of force that the human needed to perform a task through HRC. In this approach, two internal and external loops existed. The internal loop was robot specific and without requiring information regarding the task being operated; adjusted only the nonlinear behavior of the robot at the end effector such that human user could receive it based on the determined impedance model. This part was

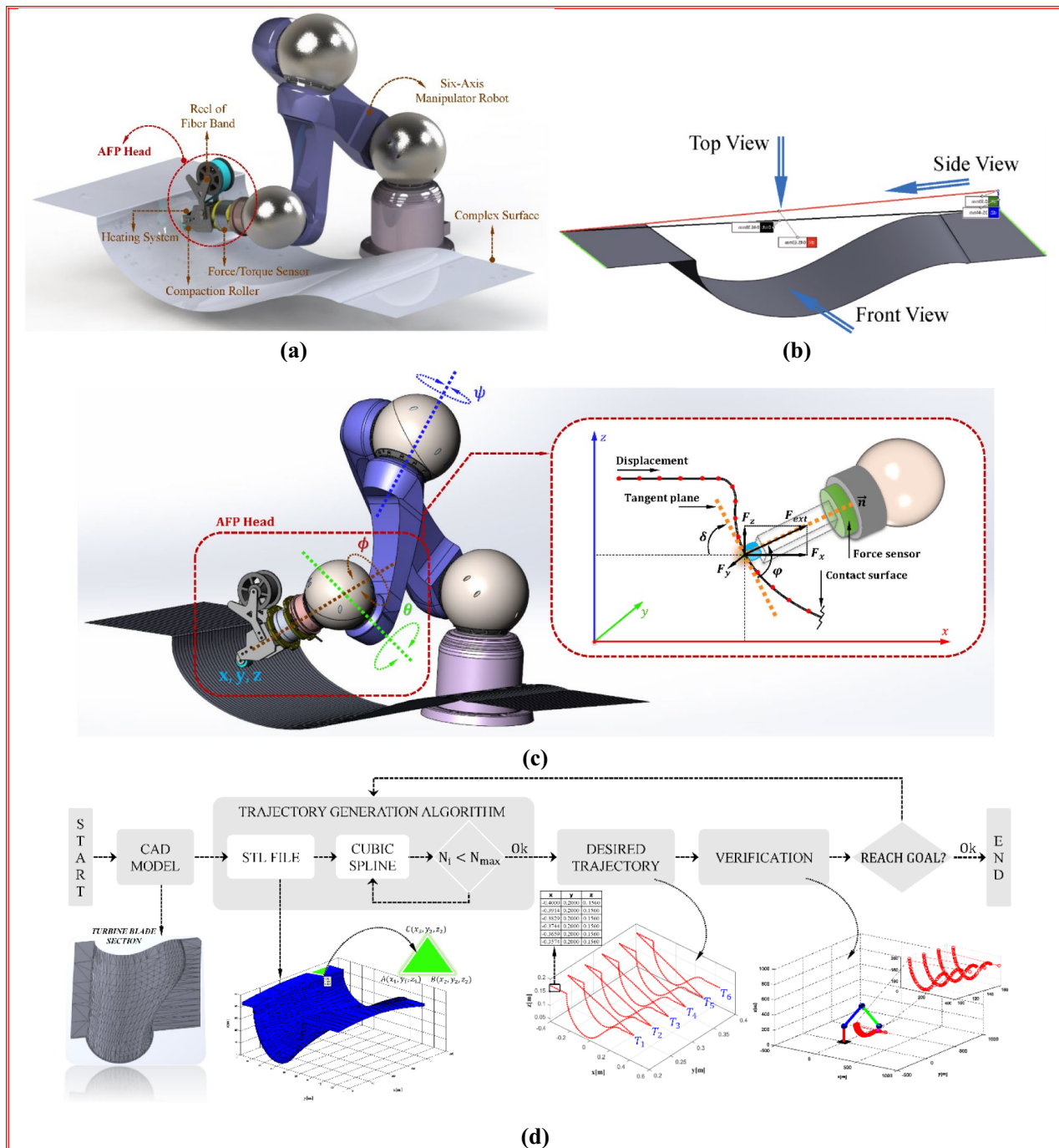


Fig. 45 Design and analysis of hybrid force/position controller for APF manipulators: **a** 3D schematic of six-axis robot with an AFP head, **(b)** the section of turbine blade section utilized in simulation

force/position control in trajectory tracking system, **(c)** the working piece tool in contact state during AFP process, and **(d)** the generation and validation method for complex trajectories [228]

trained by a NN. Besides, the external loop was task specific and based on the dynamic model of the human user and the task being assumed; adapted the dynamics of the robot with the user skills and the considered task by determining optimal parameters for impedance model. Else, the optimal impedance model parameters were found by a linear

quadratic regulator (LQR) problem which minimized human user effort and optimized the performance of the closed-loop system. Similarly, for similar approach, Li et al. [233] used integral supportive learning to solve an LQR problem which minimized route tracking error and optimized the impedance model obtained from human arm. This adaptive

impedance approach was also executed in two external and internal loops, and trained physical limitations, transition state disturbances, and dynamic uncertainties. To overpower the limitations of traditional SMC such as mechanical wear and energy inefficiency and to permit additional flexibility and smooth control action, Gaufan et al. [234] presented a novel FOSMC strategy shown in Fig. 46. The developed system was tested under dynamic instabilities characterized by abrupt or undefined change environment and revealed significant performance in disturbance rejection, trajectory tracking, and chattering drop at realistic operational circumstances.

However, Sharkawy et al. [144] used a VAC perceptron three-layer NN for HRI for the interaction of a KUKA LWR robotic manipulator and human. In this approach, NN adjusted admittance control coefficient which specifically determined the oscillation coefficient at output by considering the position and force measured at end effector. Besides, a simultaneous estimation for accuracy and speed of motion can be achieved depending on the task being undertaken by the robot. The approach was tested for the movements of end effector in a horizontal line based on the route with minimum motion jerk. To enhance HRI through whole-hand haptic rendering, Rätz et al. [235] developed a rehabilitation device specifically for upper limb by using physics-engine-driven haptic feedback. Besides, to enhance the utilized approach, the human-centered evaluation was incorporated, which evaluate usability and flexibility in clinical environment. Moreover, Sharkawy et al. [236] complemented their previous work [144]; the robot was trained in a similar

technique based on a NN of an impedance model for HRI. Furthermore, Cacace et al. [237] developed a VAC approach based on a parametric sigmoid function where hardness and oscillation values of admittance model were trained online. Besides, experimental tests showed that this approach was effective in smooth movement of end effector by human user. Meanwhile, Zhao et al. [238] presented an efficient multi-actor-critic deep deterministic policy gradient (M2ACD) algorithm improved by NURBS dual actor-critic and trajectory smoothing. To validate the performance of the developed approach, the seven-DOF SIASUN SCR3 collaborative manipulator system was used, where it revealed that the presented approach achieved better trajectory smoothness, convergence speed compared with DARC, DDPG, and TD3. In addition, Zeng et al. [239] defined a cost function with adaptive impedance control to enhance the improvement of HRI with the case study of two-way sawing and based on decreasing gradient, determined impedance model coefficients to obtain optimal interaction. These two cost functions had two parts: one was impedance control and the other one was feed-forward control which was the ultimate input to the robot as a control signal. Furthermore, Wu et al. [240] applied integral supportive learning to adjust variable impedance model parameters for HRC in moving objects. Task-dependent cost function was considered a root problem including square error of tracking and controlling effort. In 2020, they added reference route estimation to their approach and studied it in more details to complement it [240]. Gaussian mixture model (GMM) and Gaussian mixture regression (GMR) were used to model

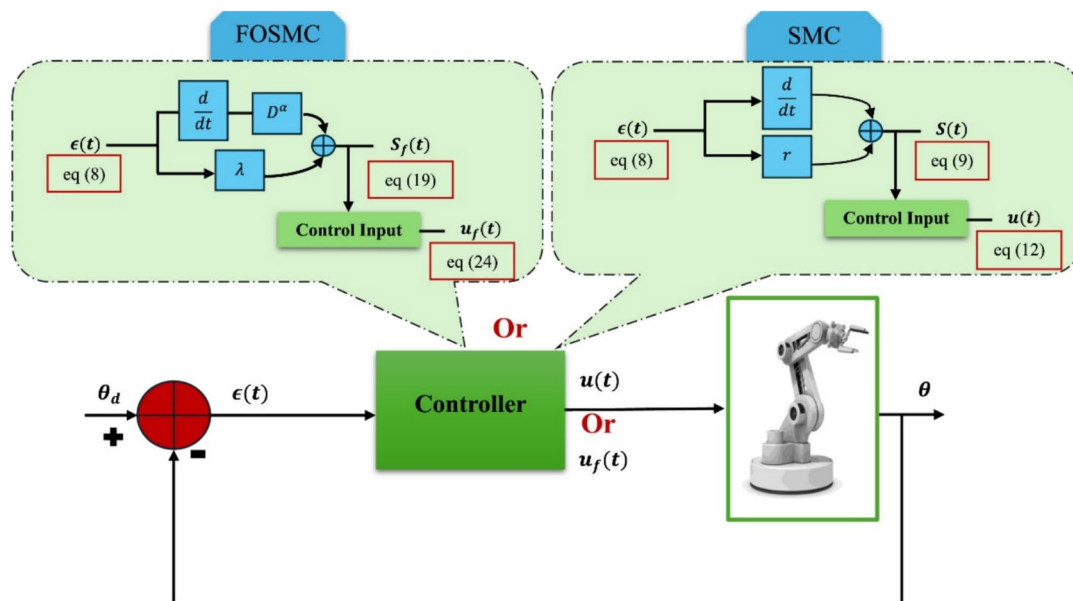


Fig. 46 The schematic block diagram of a two-link manipulator with SMC and FOSMC controller [234]

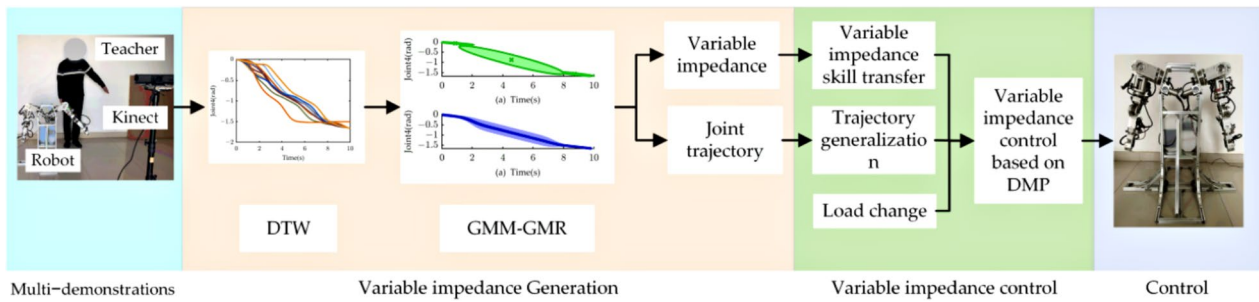


Fig. 47 The overall schematic of the implemented VIC [241]

a learning framework robot that transfer human demonstration trajectories combined with dynamic time warping (DTW) into generalized and precise motions using dynamic movement primitives (DMP), as illustrated in Fig. 47, by Zhong et al. [241]. In such system, in response to task complexity and uncertainty, the robot adapted stiffness and damping dynamically. Besides, to map trajectory variance to impedance profiles, a Softplus-based function was utilized, which in consequence ensured both accuracy and safety.

Roveda et al. [7] applied supportive learning to estimate interaction or impedance model. The estimated model was introduced to a model predictor controller (MPC), which

uses a counter-entropy method (CEM) to optimize efficiency, in order to minimize human user effort by adjusting hardness and oscillation parameters in impedance model. Moreover, Roveda et al. [25] developed a variable impedance prediction model, in which Q learning was applied to train interaction and estimate user motion intention. Besides, controlling performance was optimized based on impedance control parameters, i.e., point of adjustment and oscillation, as presented in Fig. 48.

As a further study accomplished by the same authors, Roveda et al. [242] proposed a novel concept defined as user-central adjustment, which was adjusted specifically for each human. This approach included three layers: the

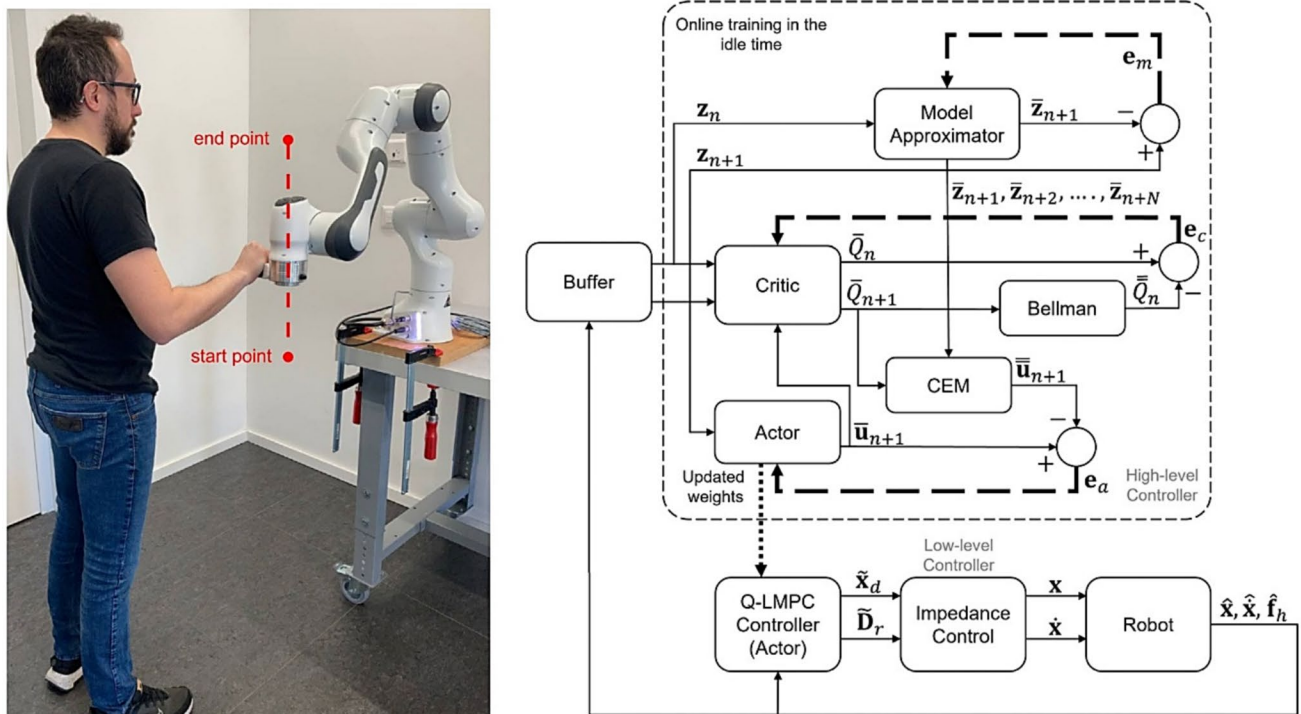


Fig. 48 Diagram of control approach and experimental setup of HRI with the idea of interactive learning (impedance model) based on Q learning method [25]

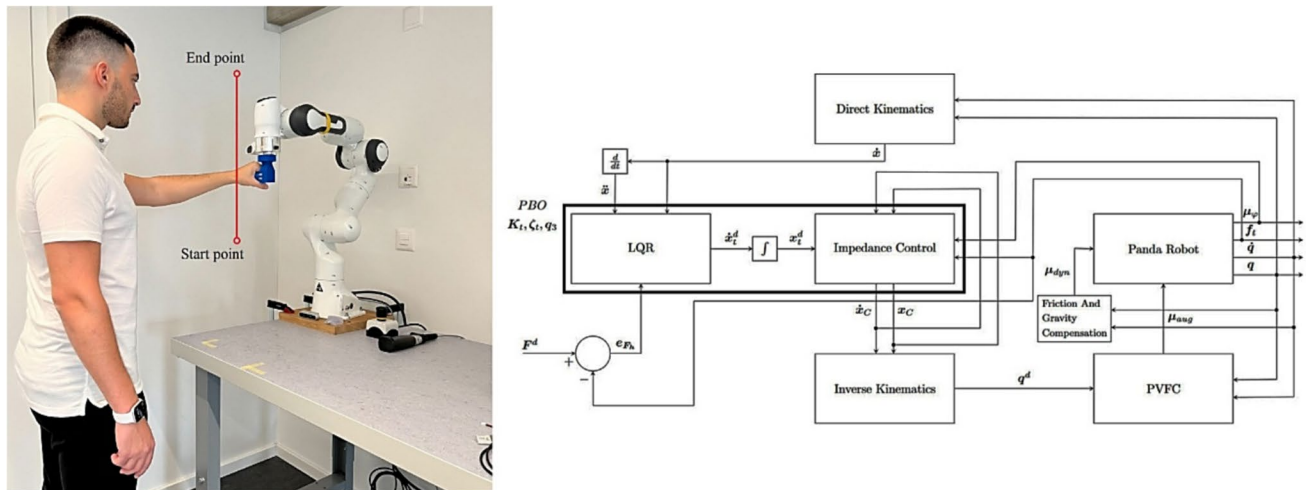
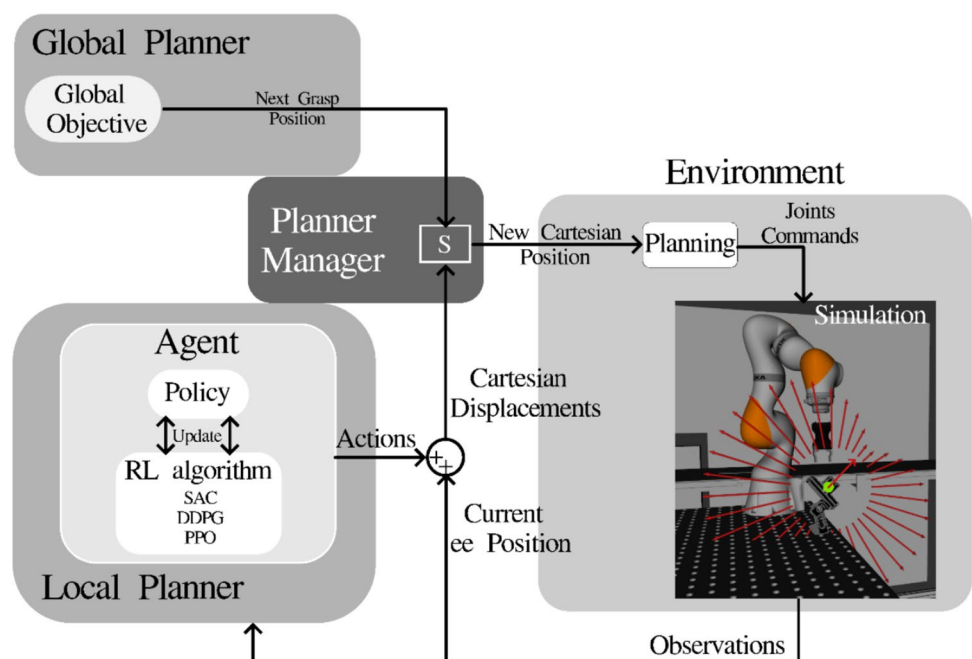


Fig. 49 Diagram of control approach and experimental setup of HRI with the idea of user-central adjustment to obtain optimized interaction [242]

first layer was speed domain control which guaranteed that the system remained inactive, the second layer was variable impedance model which managed interaction and determined route for the first layer, and the third layer estimated motion intention by defining an LQR problem, as presented in Fig. 49. Additionally, RL is commonly utilized to create flexible controllers for intricate robotic assignments. It supports in refining control settings to enhance efficiency in tasks that involve tracking force and motion [243]. Besides, the RL approach is considered one of the most interested research areas owing to its robust ability in solving robotics and control task

with high complexity and its similarity to human learning performance [244]. However, strategies for repetitive impedance learning control have been created for tasks that demand reliable performance across various iterations, guaranteeing consistent convergence of tracking errors [245, 246]. However, in robotic disassembly of flexible element, to avoid damage during extraction and reduce force by precise manipulation of robot, Tapia Sal Paz et al. [247] proposed RL base control strategy that focused on low-level control. By validating the presented method, supreme performance was revealed with accomplishing high effectiveness in dynamic environments. The

Fig. 50 The schematic of hybrid model consist of RL-based control combined with pre-planned and reactive (global and local) respectively control performance [247]



evaluation and validation processes were conducted by proximal policy optimization (PPO), deep deterministic policy gradient (DDPG), and soft actor-critic (SAC). The proposed hybrid systems of RL with pre-planned and reactive (global and local, respectively) control performance are illustrated in Fig. 50.

7 Conclusion

With increasing demands of robotics collaboration in various industries, with conserving the concepts of productivity, reliability, zero defects, and safety as main concerns, it leverages the necessity of understanding the dynamics and controls in HRI/HRC systems to obtain the maximum performance, flexibility with handling nonlinearity, and complexity under various real-time environmental conditions. This review study provides a comprehensive investigation of various VIC/VAC techniques utilized in HRC/HRI, focusing on FO, fuzzy logic, and hybrid approaches. Various industrial applications were presented to highlight the importance of cobots in recent industrial revolutions. The key findings across the sections addressed significant improvements in ensuring more adaptable, responsive, flexibility, and safety in HRI/HRC. The integration of FO derivatives, fuzzy logic, and learning-based methods significantly enhanced the performance and reduce error in HRI/HRC systems. These methods enable robots to be more adaptable, safe, and efficient in pHRI.

The improvements in VIC and VAC techniques play a key role in enhancing HRI/HRC by improving safety, adaptability, as well as responsiveness during physical interactions. These control systems allow robots to dynamically adjust their damping, stiffness, and motion parameters based on human input, such as electromyography signals, and adapt to various environmental constraints and changes and varying conditions. Moreover, hybrid methods, like those combining fuzzy systems and ML algorithms, enhance the robot performance supplementary, ensuring smoother and safer interactions, higher movability, and minimum error. For instance, fuzzy logic allows for intuitive adjustments to stiffness and damping, improving task performance in applications like object transportation and rehabilitation. On the other hand, learning-based methods, including RL and observation-based learning, optimize robot responses to human actions, reducing human effort during collaborative tasks. Additionally, using FO derivatives in VAC enhances flexibility and robustness, enabling better adaptation to dynamic environments and human motions. Overall, these developments contribute in achieving significant efficiency and safety in HRI/HRC across numerous applications. Besides, it was revealed that the dominant method in controlling interaction force relies on linear impedance

models with unknown coefficients, which are often optimized by implementing modern approaches such as learning methods. These findings contribute to an emphasized understanding of how VIC can be applied to improve HRI across different industrial and rehabilitation scenarios. In addition, by addressing limitation of linear models and robustness of nonlinear models, it is revealed that the high accuracy achieved by applying advanced approaches in nonlinear models has a direct relation with the complexity of the system.

Author contribution MRSN: conceptualization, methodology, writing—original draft, and supervision. EHA: data curation, formal analysis, figures, visualization, and writing—original draft. SS: investigation and writing—review and editing. BS: methodology, investigation, supervision, and writing—review and editing.

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Data Availability Not applicable.

Code availability Not applicable.

Declarations

Conflict of interest The authors declare no competing interests.

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